

Juniper Networks

JNCIA-Junos

Juniper Networks Certified Internet Associate

300 Interview Questions & Answers



Certification	Juniper JNCIA-Junos (JN0-105)
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How to use this guide: Each question is explained in simple, clear language with real-life analogies so you can understand and remember concepts easily. Green TIP boxes highlight exam-critical points. All 10 sections cover the complete JNCIA-Junos JN0-105 syllabus. Prepared by Akhilesh Chaudhari – Senior IT Engineer.

SECTION 01	Juniper & Junos Overview	Q1 – Q30
Junos OS · Juniper Products · CLI Basics · Software Architecture · History		
Q1	What is Juniper Networks and what does it do?	
Ans	Juniper Networks is a leading American company that makes networking hardware and software — routers, switches, firewalls, and management tools. Founded in 1996, Juniper is famous for its high-performance routers used by internet service providers (ISPs), telecom companies, and large enterprises. Think of Juniper as Cisco's biggest rival — both make the devices that run the internet.	
TIP	<i>Juniper's main product lines: Junos OS (software), MX series (routers), EX series (switches), SRX series (firewalls), QFX series (data centre switches).</i>	
Q2	What is Junos OS?	
Ans	Junos OS is Juniper Networks' operating system — the software that runs on all Juniper routers, switches, and firewalls. It is based on FreeBSD (a Unix-like system). Key fact: ONE Junos OS version runs on ALL Juniper devices — routers, switches, firewalls. This is different from Cisco, which has different OS versions (IOS, NX-OS, IOS XE) for different devices. One OS = consistent commands across all devices = easier to learn and manage.	
Q3	What makes Junos OS different from Cisco IOS?	
Ans	Key differences: Single unified OS (Junos runs on all Juniper devices; Cisco has multiple OS versions). Commit model (Junos changes are NOT applied until you type 'commit' — safe!). Junos uses set-based configuration language. CLI has two modes: operational and configuration. Junos supports rollback to any of the last 50 configurations. Junos has separate routing processes (rpd) running independently from the OS — more stable.	
Q4	What is the Junos OS software architecture?	
Ans	Junos OS runs on two separate planes: Control Plane (Routing Engine / RE): the brain — runs the OS, routing protocols, management. Runs on a dedicated CPU. Forwarding Plane (Packet Forwarding Engine / PFE): the hands — actually forwards packets at wire speed using custom hardware ASICs. They run independently. If the control plane crashes, the PFE keeps forwarding traffic (Graceful Restart). This separation makes Junos extremely stable.	
TIP	<i>The RE/PFE separation is a key Juniper architecture concept tested in JNCIA!</i>	

Q5 What is the Routing Engine (RE) in Junos?

Ans The Routing Engine is the brain of a Juniper device. It runs the Junos OS, handles: routing protocol calculations (OSPF, BGP, ISIS), management (CLI, SNMP, SSH), system processes, and populates the Routing Information Base (RIB — routing table). The RE sends forwarding information to the PFE. In high-end routers (MX series), the RE is a separate blade that can be hot-swapped without disrupting traffic.

Q6 What is the Packet Forwarding Engine (PFE)?

Ans The PFE is the muscle of a Juniper device. It receives forwarding tables from the RE and forwards packets at wire speed using hardware ASICs — no software processing for normal traffic. The PFE handles: IP forwarding, firewall filtering, traffic policing and shaping. Because it uses dedicated hardware (not software), Juniper devices can forward millions of packets per second without high CPU usage.

Q7 What are the main Juniper product families?

Ans MX Series: high-performance edge and core routers — used by ISPs and large enterprises. T/PTX Series: high-capacity core routers for carrier networks. EX Series: enterprise Ethernet switches (EX2300, EX3400, EX4300). QFX Series: high-speed data centre switches. SRX Series: security gateways/firewalls (SRX300, SRX1500, SRX5000). NFX Series: network functions virtualisation CPE. vSRX/vMX: virtual (software-only) versions running on servers or cloud.

Q8 What Juniper hardware is covered in JNCIA-Junos?

Ans JNCIA-Junos covers the J-Web and Junos CLI on all platforms, but particularly focuses on: SRX Series (branch security gateways), EX Series (enterprise switches), and the virtual vSRX/vMX platforms for lab practice. The exam is vendor-hardware agnostic — focuses on Junos OS concepts that work the same way on all Juniper devices.

Q9 What are the Juniper certification tracks?

Ans Juniper has four main certification tracks – each with three levels: JNCIA (Associate), JNCIS (Specialist), JNCIP (Professional), JNCIE (Expert). Tracks: Enterprise Routing and Switching, Service Provider Routing and Switching, Security, Cloud, DevOps, Data Centre. JNCIA-Junos (JN0-105) is the entry-level certification – foundation for all tracks. Most people start with JNCIA-Junos before specialising.

TIP JNCIA-Junos → JNCIS-ENT/SP → JNCIP → JNCIE is the typical progression.

Q10 What are the Junos CLI modes?

Ans Junos CLI has two main modes: Operational Mode (user@device>): used for monitoring, troubleshooting, and running operational commands like 'show', 'ping', 'traceroute'. Configuration Mode (user@device#): used to change device configuration. Enter with 'configure' or 'configure exclusive'. All configuration changes in Junos must be committed to take effect. This two-mode structure is fundamental to understanding Junos.

Q11 How do you access and exit configuration mode?

Ans From operational mode (>), type: configure – enters shared configuration mode (others can also edit). configure exclusive – only you can edit (others get read-only). configure private – your own copy, merged on commit. To exit configuration mode: exit – leaves config mode (returns to operational mode). quit – same as exit. exit discard – exits and discards all uncommitted changes. Ctrl+D also exits.

Q12 What is the Junos commit model and why is it powerful?

Ans The commit model means ALL configuration changes in Junos are staged in a 'candidate configuration' and do NOT take effect until you explicitly type 'commit'. Benefits: you can make multiple changes safely – nothing breaks until you commit. You can review all pending changes with 'show | compare' before committing. You can discard all changes with 'rollback 0'. If commit causes problems, you can rollback instantly. This is much safer than Cisco IOS where every command takes effect immediately.

TIP The commit model is one of Junos's most important features – memorise it for JNCIA!

Q13 What is 'commit confirmed' and when do you use it?

Ans commit confirmed [minutes] automatically rolls back the configuration if you do not confirm within the specified time (default 10 minutes). Use case: you are making a change remotely (via SSH) and worried the change might break connectivity. If it does, you cannot log back in to fix it — but 'commit confirmed' automatically undoes the change after the timeout. Confirm before timeout: 'commit' again. Very useful for remote maintenance windows.

Q14 What is rollback in Junos?

Ans Junos automatically saves the last 50 committed configurations. Rollback lets you restore any previous configuration instantly. Commands: rollback 0 (discard uncommitted changes), rollback 1 (restore previous commit), rollback 5 (restore config from 5 commits ago). After rollback, you must still commit to activate it. This is incredibly powerful — if a change causes problems, type 'rollback 1' and 'commit' to instantly restore the working configuration.

Q15 What is the difference between the candidate and active configuration?

Ans Active configuration: the configuration currently running on the device — what is actually in effect right now. Candidate configuration: your working copy where you stage changes. When you edit in configuration mode, you change the candidate, NOT the active config. When you 'commit', the candidate becomes the new active config. You can always compare them: 'show | compare' (shows differences between candidate and active). 'rollback 0' resets candidate to match active.

Q16 What is a Junos software release and how is it named?

Ans Junos version format: Major.Minor[Build-Spin]. Example: 22.4R1.10 means: 22 = year 2022, 4 = fourth release of that year, R = Release type (R=maintenance release, X=special/beta), 1 = build number, 10 = spin number. Types: R (standard maintenance), S (service release with critical fixes), F (feature release — new features). Latest versions follow the same pattern. Always check Juniper's release notes before upgrading.

Q17 How do you upgrade Junos OS?

Ans Upgrade steps: Download new Junos package from Juniper support portal. Copy to device: 'scp junos-install-xxx.tgz user@device:/var/tmp/'. Install: request system software add /var/tmp/junos-install-xxx.tgz. Reboot: request system reboot. Verify: show version. Always do: backup current config before upgrade, verify enough disk space (request system storage cleanup), test in lab before production. Junos supports ISSU (In-Service Software Upgrade) on high-end platforms – zero downtime upgrade.

Q18 What is the Junos file system and important directories?

Ans Key Junos directories: /var/tmp – temporary files (store software packages here for installation). /var/log – system and event log files. /var/db/config – configuration files (juniper.conf = active config, juniper.conf.1 through .49 = rollback copies). /cf – compact flash storage. /boot – kernel and boot files. /etc – system configuration. View storage: 'show system storage'. Clean up: 'request system storage cleanup'.

Q19 What is J-Web?

Ans J-Web is Junos's web-based graphical management interface – access it by opening a browser and navigating to the device's IP address (http or https). It provides: dashboard (device health, interfaces, BGP status), configuration wizard, monitoring (traffic graphs, logs), software upgrade. J-Web is useful for beginners or for quick checks without needing CLI knowledge. However, most Juniper engineers primarily use CLI for advanced configurations.

Q20 What are Junos user accounts and privilege levels?

Ans Junos supports multiple users with different privilege levels (login classes). Built-in classes: super-user (all permissions – equivalent to root), operator (can view, clear, and reset – but cannot configure), read-only (show commands only), unauthorized (no access). Create user: set system login user akhilesh class super-user authentication plain-text-password. Best practice: avoid using root for daily work – create named accounts for accountability and audit trails.

Q21 How do you configure SSH access on Junos?

Ans Enable SSH: set system services ssh. Set root authentication (required): set system root-authentication plain-text-password (then type password). Generate SSH keys (optional but recommended): request security generate-key-pair. Allow specific hosts: set system services ssh allowed-hosts 192.168.1.0/24. Disable Telnet for security: delete system services telnet. Commit changes. Verify SSH is listening: show system services.

Q22 What is the Junos rescue configuration?

Ans The rescue configuration is a minimal, known-good configuration saved separately from the normal configuration. Used when you accidentally commit a configuration that locks you out of the device (broken SSH config, wrong IP). Load rescue config: request system configuration rescue rollback. Create/save rescue config: request system configuration rescue save. The rescue config typically contains just: hostname, management IP, SSH, and root password.

Q23 What is chassis cluster in Junos (SRX)?

Ans Chassis cluster connects two SRX firewalls as an active-passive (or active-active) high-availability pair. One node is Active (handles traffic), one is Passive (standby, ready to take over). State synchronisation: both nodes sync session tables, routing, and configuration over dedicated cluster control links. Failover: if Active node fails, Passive takes over in seconds. Users notice minimal disruption. Used in data centres and critical sites requiring 24/7 uptime.

Q24 What is the Junos event policy?

Ans Event policy (event-options) allows the router to automatically execute actions when specific system events occur – without human intervention. Example: when an interface goes down, automatically send an email alert and log a message. Example: when BGP session resets, run a diagnostic script. Configure: set event-options policy IF_DOWN events SNMP_TRAP_LINK_DOWN. Very powerful for automation and proactive network management.

Q25 What is NETCONF in Junos context?

Ans NETCONF is an open standard protocol for programmatically managing network devices – used for automation and SDN. Junos has native NETCONF support. It uses XML to send and receive configuration data. Automation tools (Ansible, Python with PyEZ library) use NETCONF to configure Junos devices remotely without manual CLI. Enable: set system services netconf ssh. Juniper's PyEZ Python library makes Junos automation very simple.

Q26 What is the Junos PyEZ library?

Ans PyEZ is Juniper's Python library for automating Junos devices. It provides simple Python functions to: connect to devices, get operational data (show commands), push configurations, retrieve facts (hostname, version, uptime). Install: pip install junos-eznc. Example: `device = Device(host='10.0.0.1', user='admin', password='pass');` `device.open(); facts = device.facts`. Used by network engineers for automated monitoring, configuration management, and bulk changes.

Q27 What is graceful restart in Junos?

Ans Graceful Restart allows a Juniper device to keep forwarding traffic (using the PFE) even when the Routing Engine restarts or the routing daemon crashes. The PFE continues to forward packets using its cached forwarding table. Neighbouring routers are told to hold their routes ('helper mode') while the RE restarts and rebuilds its routing table. Users experience no service interruption. This is only possible because of the RE/PFE separation in Junos architecture.

Q28 What is NSR (Non-Stop Routing) in Junos?

Ans NSR allows routing protocol sessions (BGP, OSPF, ISIS) to continue without interruption even when the primary Routing Engine fails and a backup RE takes over (on dual-RE platforms like MX480). The backup RE maintains a live copy of all routing protocol state. When primary RE fails: backup RE instantly becomes primary without neighbours noticing. Sessions do not reset. No traffic disruption. Requires dual RE hardware and NSR configuration.

Q29 What is the Junos management interface?

Ans The management interface (fxp0 on most Juniper devices, me0 on some) is a dedicated out-of-band management port. It connects to the management network for SSH, SNMP, Syslog – completely separate from traffic interfaces. Best practice: use fxp0 for all management access – if a traffic interface has issues, you can still access the device via fxp0. Configure: set interfaces fxp0 unit 0 family inet address 192.168.1.1/24.

Q30 What is the difference between JNCIA, JNCIS, JNCIP, and JNCIE?

Ans JNCIA (Associate): entry level – covers Junos basics, interface config, routing fundamentals. Exam: JN0-105 (JNCIA-Junos). JNCIS (Specialist): intermediate – deeper routing, switching, or security. Example: JNCIS-ENT covers OSPF, BGP, VLANs in depth. JNCIP (Professional): advanced – complex network design and troubleshooting. JNCIE (Expert): highest – includes both written and 8-hour practical lab exam. Each level requires passing the exam (no prerequisite exams for JNCIA).

SECTION 02

Junos CLI & Configuration Basics

Q31 – Q60

CLI Navigation · Show Commands · Configuration Hierarchy · Editing · Verification

Q31 What is the Junos configuration hierarchy?

Ans Junos configuration is organised in a tree-like hierarchy using stanzas (sections). Top level: system, interfaces, routing-options, protocols, security, firewall, etc. Each level uses indented brackets. Example: interfaces { ge-0/0/0 { unit 0 { family inet { address 192.168.1.1/24; } } } }. You navigate the hierarchy in set commands: set interfaces ge-0/0/0 unit 0 family inet address 192.168.1.1/24. Very logical and easy to understand once you grasp the tree structure.

TIP *Understanding the configuration hierarchy is fundamental to all Junos configuration!*

Q32 How do you navigate the configuration hierarchy?

Ans In configuration mode, use 'edit' to go deeper into a hierarchy level: edit interfaces ge-0/0/0 (now you are inside the ge-0/0/0 context). Then: edit unit 0 □ edit family inet. From here, shorter commands work. Go back up: up (one level), top (back to root). Show current position: show (shows config from current level down). The hierarchy navigation is like navigating folders in a file system – cd to go in, cd.. to go back.

Q33 What is the 'set' command in Junos configuration?

Ans The 'set' command adds or changes a configuration parameter. Syntax: set [hierarchy path] [parameter] [value]. Examples: set system hostname JuniperRouter1, set interfaces ge-0/0/0 unit 0 family inet address 10.1.1.1/24, set protocols ospf area 0 interface ge-0/0/0.0. You can type the full path from the configuration root, or navigate into the hierarchy first and use shorter commands. Remember: nothing takes effect until 'commit'!

Q34 What is the 'delete' command in Junos?

Ans The 'delete' command removes a configuration item. Examples: delete interfaces ge-0/0/1 (removes entire interface config), delete protocols ospf area 0 interface ge-0/0/1.0 (removes just that interface from OSPF). Be careful: 'delete' at a high level removes everything below that level too. Always verify what you are deleting with 'show [path]' first. After deleting, you still must 'commit' for the change to take effect.

Q35 What is the 'show' command in Junos operational mode?

Ans Show commands display current device state without changing anything – safe to run anytime. Key show commands: show version (Junos version, hardware model), show interfaces (interface status and stats), show route (routing table), show bgp summary (BGP peer status), show ospf neighbor (OSPF neighbour status), show system uptime (device uptime), show log messages (system log), show chassis hardware (hardware inventory). Output is structured and readable.

Q36 What does 'show | compare' do in Junos?

Ans In configuration mode, 'show | compare' displays the difference between your current candidate configuration and the active running configuration. Lines starting with + will be added, lines starting with - will be removed. This is incredibly useful for reviewing your changes before committing – a final sanity check. Example: you change an IP address – 'show | compare' shows the old IP with '-' and new IP with '+'. Always use this before 'commit'!

Q37 What are output modifiers (pipes) in Junos CLI?

Ans Pipes (|) filter or modify show command output – just like Linux. Key modifiers: | match [text] – show only lines containing text (like grep). | except [text] – exclude lines with text. | count – count number of lines. | find [text] – jump to first occurrence. | display set – show config in 'set' command format. | no-more – disable pagination (all output at once). Example: show route | match 192.168.1 – shows only routes containing that IP.

Q38 What is 'show | display set'?

Ans This displays the active configuration in 'set' command format – each configuration item shown as a complete 'set' command. Useful for: understanding what commands were used to create the current config, copying commands to another device, creating a change script. Example: show interfaces | display set shows output like: set interfaces ge-0/0/0 unit 0 family inet address 10.1.1.1/24. Very useful for documentation and migration.

Q39 How do you search for text in Junos configuration?

Ans Two ways to search: In configuration mode: `show [path] | match [search-term]` – shows config lines matching the term. In configuration mode: `show | display set | match [term]` – search all set commands. Operational mode: `show configuration | match [term]` – search entire config. Example: `show configuration | match ospf` – shows all config lines mentioning ospf. Helps quickly find where something is configured in a complex config.

Q40 What is the Junos 'request' command?

Ans Request commands trigger operational actions – system operations beyond just viewing information. Key request commands: `request system reboot` (reboot the device), `request system halt` (graceful shutdown), `request system software add [file]` (install new Junos), `request system storage cleanup` (remove old files), `request system snapshot` (backup config and OS to storage media), `request security pki generate-key-pair` (generate SSH/TLS keys).

Q41 How do you copy files to and from a Junos device?

Ans Junos supports standard file transfer methods: SCP (Secure Copy – most common): `scp local-file user@device:/var/tmp/` from your laptop. From device: `file copy [source] [destination]`. FTP (if enabled): `set system services ftp`. USB: plug in USB drive and use `file list /dev/da0` and `file copy`. Internal copy: `file copy /var/tmp/file1 /var/log/`. The 'file' command in Junos CLI handles most file operations. Always use SCP (encrypted) rather than FTP (plaintext).

Q42 What is the 'top' command in Junos configuration mode?

Ans The 'top' command returns you to the top (root) of the configuration hierarchy from wherever you are currently. Useful when you have navigated deep into a hierarchy: `edit interfaces ge-0/0/0 unit 0 family inet` □ then type 'top' □ instantly back at root. Without 'top', you would need multiple 'up' commands or 'exit' and re-enter config mode. Saves time when switching between different parts of the configuration.

Q43 What is auto-completion in Junos CLI?

Ans Junos CLI supports Tab completion – press Tab to complete a command or show available options. Press Space for abbreviation completion. Press ? to show all valid options at any point. Example: type 'sh' and press Tab $\square$ completes to 'show'. Type 'show int' and press Tab $\square$ shows all interface-related show commands. This makes CLI much faster – you do not need to type full command names. Essential skill for working efficiently in Junos.

Q44 How do you load a configuration from a file?

Ans In configuration mode: load [options] [filename]. Options: load override [file] – completely replace current candidate with file. load merge [file] – merge file contents into existing candidate. load replace [file] – file lines with replace: tag replace matching sections. load patch [file] – apply unified diff patch. Load from URL: load merge [url] terminal (paste config). Example: load override /var/tmp/backup-config.txt. Useful for restoring configs or applying scripted changes.

Q45 What is the Junos 'save' configuration command?

Ans In configuration mode: save [filename] saves the current candidate configuration to a file. Example: save /var/tmp/backup-before-change.txt. This creates a text file of the configuration in the Junos hierarchy format. You can later restore with 'load override [file]'. Best practice: always save a backup before making major changes. Also useful for: moving config to another device, creating configuration templates, auditing.

Q46 How do you view the current active configuration in Junos?

Ans show configuration – shows the complete active running configuration in hierarchy format. Show specific section: show configuration interfaces (just interface config), show configuration protocols ospf (just OSPF config). Show in set format: show configuration | display set. Compare operational data: show version (version and model), show system configuration (system section). Show recently committed changes: show system commit (list of last commits with timestamp and user).

Q47 What does 'show system commit' display?

Ans show system commit shows the history of the last commits – who committed, when, and from which terminal. Output example: 0 [date/time] by user via cli, 1 [date/time] by admin via ssh. This is your audit trail. Very useful for troubleshooting: 'the network broke at 3 PM – show system commit tells me someone made a change at 2:58 PM from this IP address.' Helps identify the cause of issues and who made changes.

Q48 What is the 'annotate' command in Junos configuration?

Ans The 'annotate' command adds a comment to a configuration element. Example: `annotate interfaces ge-0/0/0 'Connected to Core Switch Floor 3'`. Comments appear in 'show configuration' output preceded by `/* */`. Useful for documentation inline with the configuration – team members can see notes explaining why something was configured a certain way. More useful than external documentation because it lives with the config.

Q49 How do you rename or deactivate a configuration element?

Ans Rename: `rename interfaces ge-0/0/0 to ge-0/0/1` (renames interface config block). Deactivate: `deactivate interfaces ge-0/0/1` – marks the element as inactive (deactivated sections are enclosed in `/* */` and ignored but preserved). To re-enable: `activate interfaces ge-0/0/1`. Useful for temporarily disabling a feature without deleting its configuration – you can quickly re-activate it if needed. Deactivated elements show up in 'show configuration' but are not applied.

Q50 What is the Junos configuration groups feature?

Ans Configuration groups allow you to define a set of configuration that can be applied to multiple elements at once – like a template. Define group: `set groups UPLINK-TEMPLATE interfaces mtu 9000`. Apply to all matching interfaces: `set apply-groups UPLINK-TEMPLATE`. The angle brackets `<>` are wildcards. This is powerful for applying consistent settings (MTU, authentication keys, routing policies) across many interfaces or protocols without copy-pasting. Verify: `show configuration groups`.

Q51 What is 'show interfaces terse' vs 'show interfaces'?

Ans show interfaces — detailed output for all interfaces including counters, errors, physical layer info. Often very long. show interfaces terse — compact one-line summary per interface showing: interface name, status (up/down), link (up/down), and IP address. Much easier to read at a glance. show interfaces [interface-name] — detailed info for just one interface. For quick status check, use 'terse'. For troubleshooting errors, use detailed output.

Q52 How do you clear interface counters in Junos?

Ans clear interfaces statistics [interface-name] — resets all counters for a specific interface to zero. clear interfaces statistics all — resets counters on all interfaces. Why reset? When troubleshooting, you want to see errors accumulating from a specific point in time, not from when the device booted. Reset counters, wait a few minutes, then check again to see if new errors are appearing. No config change — safe to run anytime.

Q53 What is 'show log messages' in Junos?

Ans show log messages — displays the system message log — events logged by the device like interface up/down, protocol changes, authentication failures, hardware events. show log messages | last 50 — show last 50 log entries. show log messages | match error — show only error messages. Real-time monitoring: monitor start messages — live streaming of new log messages as they appear. Ctrl+C to stop. Essential for troubleshooting — most problems leave clues in the log.

Q54 What is the Junos traceroute command?

Ans traceroute [destination] traces the route packets take to a destination, showing each router (hop) along the path with response time. In Junos: traceroute 8.8.8.8. Options: traceroute 8.8.8.8 source 10.1.1.1 (use specific source IP), traceroute 8.8.8.8 routing-instance MGMT (use specific routing instance). * * * means that router does not respond to traceroute probes — does not necessarily mean a problem. Useful for pinpointing where routing fails.

Q55 What is 'ping' in Junos and its useful options?

Ans ping [destination] — tests basic IP connectivity. Useful options: ping 8.8.8.8 count 100 (send 100 pings), ping 8.8.8.8 size 1400 (test with large packets), ping 8.8.8.8 source 10.1.1.1 (use specific source IP), ping 8.8.8.8 rapid (fast ping, shows summary at end), ping 8.8.8.8 routing-instance VPN-A (test connectivity within a specific VRF/routing instance). Ctrl+C stops a running ping.

Q56 What is 'show route' in Junos?

Ans show route — displays the routing table (all routes the device knows). Key output: destination prefix, route preference (lower = preferred), metric, next-hop IP and interface. show route [prefix] — details for a specific route. show route detail — verbose output with all attributes. show route protocol ospf — only OSPF-learned routes. show route forwarding-table — the actual forwarding table used by PFE (may differ from routing table when multiple protocols know same route).

Q57 What is the Junos 'request' vs 'show' command difference?

Ans Show commands: read-only, display current state — safe to run any time without risk. Examples: show interfaces, show route. Request commands: trigger active operations that change state or cause actions. Examples: request system reboot, request clear bgp neighbor all, request system storage cleanup. Always be careful with 'request' commands in production — especially reboot and clear commands. 'Show' is diagnostic; 'request' is operational action.

Q58 How do you test a configuration change before fully committing?

Ans Junos provides several safety mechanisms: commit check — validates syntax of candidate config without activating it (catches typos, missing references). show | compare — review all pending changes before commit. commit confirmed 5 — commits but automatically rolls back in 5 minutes if you do not confirm. Use these in order: 1) show | compare (verify changes look correct), 2) commit check (syntax valid?), 3) commit confirmed 10 (safe commit with auto-rollback option).

Q59 What is the difference between 'exit' and 'quit' in Junos?

Ans In configuration mode: exit — exits current level (if at root, returns to operational mode). If you have uncommitted changes, Junos warns you. quit — same as exit at root level — exits configuration mode. exit discard — immediately exits and discards ALL uncommitted changes without prompting. Very useful if you realise your changes are wrong and want to abandon them all at once. Ctrl+D also exits configuration mode.

Q60 How do you verify a Junos configuration is syntactically correct?

Ans `commit check` — runs syntax and semantic validation on the candidate configuration without activating it. It checks: all referenced objects exist (e.g. routing policy referenced in BGP actually exists), syntax is correct, no conflicting statements. If errors found: displays specific error messages with the location. Fix errors and run `commit check` again. Only when `commit check` passes should you proceed to `'commit'`. This is the recommended workflow for all production changes.

SECTION 03

Interface Configuration

Q61 – Q90

*Physical Interfaces · Logical Units · IPv4/IPv6 · Loopback · Monitoring · Families***Q61** What is a physical interface in Junos?

Ans A physical interface is a real hardware port on the device. Junos interface naming format: [type]-[FPC/PIC/port]. Example: ge-0/0/0 means: ge = Gigabit Ethernet, 0 = FPC slot 0, 0 = PIC slot 0, 0 = port 0. Other types: et = 100GbE, xe = 10GbE, fe = FastEthernet, ae = aggregated Ethernet (LAG), lo0 = loopback, fxp0 = management. The naming is logical – once you know the pattern, you can instantly identify any interface.

TIP *Interface naming is tested in JNCIA. Memorise: ge=1GbE, xe=10GbE, et=100GbE.*

Q62 What is a logical interface (unit) in Junos?

Ans In Junos, every physical interface has one or more logical interfaces called 'units'. A unit is where you apply IP addresses and other Layer 3 settings. For standard Ethernet: unit 0 (single unit, like a regular interface). For sub-interfaces (VLANs/tunnels): unit 100, unit 200, etc. (one per VLAN, with 802.1Q encapsulation). Physical = hardware, Unit = logical overlay. You MUST configure a unit before assigning an IP address.

Q63 How do you configure a basic IP address on a Junos interface?

Ans Configuration: set interfaces ge-0/0/0 unit 0 family inet address 192.168.1.1/24. No need for 'no shutdown' – Junos interfaces are enabled by default if not explicitly disabled. To disable: set interfaces ge-0/0/0 disable. To verify: show interfaces ge-0/0/0 terse (shows IP and status). Junos uses 'family inet' for IPv4 and 'family inet6' for IPv6. You can add multiple IP addresses on the same unit.

Q64 What is 'family inet' in Junos interface configuration?

Ans family inet enables IPv4 on a logical interface. You add IP addresses under 'family inet'. Without configuring 'family inet', the interface cannot process IPv4 traffic. Similarly, 'family inet6' for IPv6, 'family mpls' for MPLS label processing, 'family ethernet-switching' for Layer 2 switching, 'family iso' for IS-IS routing. A single unit can have multiple families – enabling multiple protocols on the same logical interface simultaneously.

Q65 How do you assign an IPv6 address in Junos?

Ans set interfaces ge-0/0/0 unit 0 family inet6 address 2001:db8::1/64. Junos supports dual-stack – you can have both IPv4 and IPv6 on the same interface simultaneously: set interfaces ge-0/0/0 unit 0 family inet address 192.168.1.1/24 AND set interfaces ge-0/0/0 unit 0 family inet6 address 2001:db8::1/64. Verify: show interfaces ge-0/0/0 detail | match inet6. Junos handles IPv6 natively without special configuration beyond the address assignment.

Q66 What is a loopback interface (lo0) in Junos?

Ans lo0 is the loopback interface in Junos – a virtual interface that is always up as long as the device is running. Key uses: Router ID (OSPF and BGP use the loopback IP as stable identifier), management reachability (loopback IP never goes down even if physical interfaces do), testing and verification. Configure: set interfaces lo0 unit 0 family inet address 10.0.0.1/32. Always use /32 mask for loopback IPs. Unlike Cisco, Junos loopback is NOT lo0 self by default – you must explicitly configure it.

Q67 How do you configure a sub-interface (VLAN tagging) in Junos?

Ans For 802.1Q VLAN sub-interfaces: set interfaces ge-0/0/0 vlan-tagging (enable tagging on physical interface). Then per VLAN: set interfaces ge-0/0/0 unit 100 vlan-id 100 (VLAN tag = 100). set interfaces ge-0/0/0 unit 100 family inet address 192.168.100.1/24. Each unit has its own VLAN ID and IP address. This is used for Router-on-a-Stick inter-VLAN routing or when connecting to a trunk port from a switch.

Q68 What is interface MTU in Junos and how do you change it?

Ans MTU (Maximum Transmission Unit) is the largest packet size an interface will forward without fragmentation. Default Ethernet MTU: 1500 bytes. To support jumbo frames (data centres, storage traffic): set interfaces ge-0/0/0 mtu 9000. For MPLS (labels add overhead): typically set 1600 bytes to avoid fragmentation. Mismatched MTU between two connected devices causes: large packets dropped, OSPF adjacency stuck in EXSTART/EXCHANGE, slow file transfers. Check: show interfaces detail | match MTU.

Q69 How do you configure interface speed and duplex in Junos?

Ans On copper Ethernet interfaces: set interfaces ge-0/0/0 speed 1g (or 100m, 10m). set interfaces ge-0/0/0 duplex full (or half). For auto-negotiation (default and recommended): set interfaces ge-0/0/0 auto-negotiation. Best practice: let both sides auto-negotiate unless connecting to legacy equipment that requires fixed settings. Duplex mismatch (one side full, other half) causes interface errors and very slow performance. Verify: show interfaces ge-0/0/0 detail | match speed.

Q70 What is interface dampening in Junos?

Ans Interface dampening suppresses a flapping interface (one that repeatedly goes up and down) from affecting the routing table and disrupting the network. Each time the interface goes down, it accumulates a penalty. When penalty exceeds a threshold, the interface is suppressed (not advertised to routing protocols). Penalty gradually decays over time. When interface is stable, it is re-enabled. Configure: set interfaces ge-0/0/0 damping enable. Prevents routing table instability from a problematic interface.

Q71 What is 'hold-time' on a Junos interface?

Ans Hold-time prevents brief interface flaps from triggering routing protocol changes. set interfaces ge-0/0/0 hold-time up 2000 down 0 means: wait 2000 milliseconds (2 seconds) after interface comes up before notifying routing protocols (prevents routing changes from brief link bounces). Down hold-time: 0 means immediately notify when interface goes down. This is a stability mechanism — common in environments with occasionally unreliable physical links.

Q72 How do you configure a static ARP entry in Junos?

Ans Static ARP maps a specific IP address to a specific MAC address permanently — used when ARP needs to be controlled (security, certain server deployments). Configuration: set interfaces ge-0/0/0 unit 0 family inet address 192.168.1.1/24 arp 192.168.1.100 mac 00:11:22:33:44:55. This tells Junos: 'to reach 192.168.1.100, always use MAC 00:11:22:33:44:55, do not do ARP.' Verify: show arp no-resolve. Useful for preventing ARP spoofing on critical servers.

Q73 What is interface description in Junos and why use it?

Ans Interface descriptions document what a port is connected to – essential for network management. `set interfaces ge-0/0/0 description 'To CoreSwitch-A Port 1/1'`. `show interfaces terse` displays descriptions next to each interface. Benefits: quick identification during troubleshooting ('which physical port is this?'), documentation, reduces errors when making changes. Best practice: always set descriptions on all connected interfaces using a consistent format: 'To [device] [port] – [purpose]'.

Q74 What is the difference between 'up' and 'down' interface states in Junos?

Ans Interface state has two components: Physical (link) state: is the cable physically connected? Is there a signal? 'up' = cable connected, signal present. 'down' = no cable or no signal. Protocol state: is Layer 2/3 working? 'up' = the interface protocol is functioning. 'down' = configured as 'disable', or no IP address, or LACP negotiation failing. In `show interfaces`: 'ge-0/0/0 up up' = both physical and protocol working. 'ge-0/0/0 up down' = cable connected but Layer 2/3 issue.

Q75 What does 'admin down' mean in Junos?

Ans Admin down means the interface has been manually disabled using the 'disable' command: `set interfaces ge-0/0/0 disable`. The interface shows as 'down down' in `show interfaces terse`, but with 'Administratively down' in the detailed output. To bring it back up: `delete interfaces ge-0/0/0 disable`. Unlike Cisco IOS where interfaces are shut by default on routers, Junos interfaces are UP by default – you explicitly disable them when needed.

Q76 What is a tunnel interface in Junos?

Ans A tunnel interface is a virtual interface that encapsulates packets inside another protocol for transport. Types: `ip-0/x/x` (IP-IP tunnel), `gr-0/x/x` (GRE tunnel), `ip6tnl` (IPv6 tunnel), `st0` (secure tunnel for IPsec VPN). Configure GRE: `set interfaces gr-0/0/0 unit 0 tunnel source [local-IP] tunnel destination [remote-IP] family inet address [tunnel-IP/mask]`. Tunnel interfaces enable: routing across different protocol domains, creating point-to-point links over routed networks.

Q77 What is aggregated Ethernet (ae) interface in Junos?

Ans Aggregated Ethernet (ae) is a Link Aggregation Group (LAG) – multiple physical interfaces bundled as one logical interface for higher bandwidth and redundancy. Configure: set chassis aggregated-devices ethernet device-count 1 (reserve ae interfaces). set interfaces ge-0/0/0 ether-options 802.3ad ae0 (add physical member). set interfaces ae0 aggregated-ether-options minimum-links 1 (min links to stay up). set interfaces ae0 unit 0 family inet address 10.1.1.1/30. LACP: add lacp active or passive for negotiation.

Q78 How do you check interface error counters in Junos?

Ans show interfaces ge-0/0/0 detail – shows all counters including: Input errors (CRC errors from bad cable, runts, giants), Output errors (queue drops from congestion), Carrier transitions (how many times link went up/down). show interfaces ge-0/0/0 statistics – same with running stats. Non-zero error counters indicate problems: CRC errors = cable/NIC issue. Input errors + drops = congestion. Many carrier transitions = unstable physical link. Clear counters to baseline: clear interfaces statistics ge-0/0/0.

Q79 What is IP unnumbered interface in Junos?

Ans An IP unnumbered interface borrows the IP address of another interface (usually lo0) instead of having its own unique IP. Saves IP addresses on point-to-point links. Configure: set interfaces ge-0/0/0 unit 0 family inet unnumbered-address lo0.0. The interface uses the loopback's IP as its address. Routing protocols can still run over it. Common on serial or point-to-point Ethernet links to service provider networks where you want to save public IP addresses.

Q80 How do you monitor interface traffic in real time?

Ans show interfaces ge-0/0/0 – shows current traffic rate and counters. For live monitoring: monitor interface ge-0/0/0 – updates stats every second in real time (press 'q' to quit). monitor interfaces (plural) – shows all interfaces updating. This is Junos's built-in real-time traffic monitor. For historical bandwidth graphs: use SNMP with a NMS like Cacti, Grafana, or PRTG to poll interface counters and plot graphs over time.

Q81 What is family ethernet-switching in Junos?

Ans family ethernet-switching enables Layer 2 switching on a Junos interface (used on EX switches and SRX in switching mode). When configured, the interface becomes a switch port (access or trunk) rather than a Layer 3 routed port. set interfaces ge-0/0/0 unit 0 family ethernet-switching interface-mode access vlan members VLAN10. This configures the port as an access port in VLAN 10. For trunk: interface-mode trunk vlan members [VLAN-list].

Q82 What is IRB (Integrated Routing and Bridging) interface in Junos?

Ans IRB is Junos's equivalent of a Cisco SVI (Switch Virtual Interface) – a virtual Layer 3 interface for a VLAN that enables routing between VLANs on an EX switch or SRX. Configure: set interfaces irb unit 10 family inet address 192.168.10.1/24 (gateway for VLAN 10). Devices in VLAN 10 use 192.168.10.1 as their default gateway. Then enable routing: set vlans VLAN10 l3-interface irb.10. Much cleaner than Router-on-a-Stick.

Q83 What is the difference between a routed interface and a switched interface?

Ans Routed interface (family inet/inet6): operates at Layer 3, has an IP address, participates in routing protocols. Forwards packets based on IP routing table. Used on router-to-router or router-to-server links. Switched interface (family ethernet-switching): operates at Layer 2, forwards frames based on MAC address table. Used on switch-to-PC or switch-to-switch links. Junos EX switches can have both: switched ports for users, and routed IRB interfaces for inter-VLAN routing.

Q84 What is the significance of '.0' after an interface name?

Ans In Junos, the '.0' (or '.100', '.200') represents the unit number of a logical interface. ge-0/0/0.0 = physical interface ge-0/0/0, logical unit 0. For most point-to-point links, only unit 0 is used. For 802.1Q sub-interfaces: ge-0/0/0.100, ge-0/0/0.200 – each unit is a different VLAN. In routing protocol and policy configurations, you reference the full logical interface including the unit: ospf area 0 interface ge-0/0/0.0. This is a common source of confusion for those coming from Cisco IOS.

Q85 How do you configure an interface for DHCP client in Junos?

Ans To get IP address from DHCP server: set interfaces ge-0/0/0 unit 0 family inet dhcp-client. Commit. The interface will send DHCP Discover and configure itself with the received IP. Verify: show interfaces ge-0/0/0 detail (shows 'DHCP' under address). Useful for branch devices or management interfaces getting IP from DHCP server. Can also do: show dhcp client binding (see lease information).

Q86 What is a virtual interface in Junos (VRRP)?

Ans VRRP (Virtual Router Redundancy Protocol) creates a virtual IP address shared between two Junos devices for gateway redundancy. Configure: set interfaces ge-0/0/0 unit 0 family inet address 10.1.1.2/24 vrrp-group 1 virtual-address 10.1.1.1. The virtual-address (10.1.1.1) is what clients use as gateway. One router is Master (handles traffic), one is Backup. If Master fails, Backup takes over the virtual IP. Junos supports both VRRP (open standard) and its enhanced version.

Q87 How do you verify IP connectivity using Junos tools?

Ans Step-by-step verification: show interfaces terse (check interface status and IPs). ping [next-hop-IP] (test Layer 3 reachability to directly connected device). show arp (verify ARP resolution working). ping [distant-IP] routing-instance [vrf-name] (test through specific routing instance). traceroute [destination] (trace path if ping fails). show route [destination] (check routing table has correct route). This layered approach quickly isolates where connectivity breaks.

Q88 What is LACP (Link Aggregation Control Protocol) in Junos?

Ans LACP (IEEE 802.3ad) negotiates and manages link aggregation groups automatically. Active mode: device initiates LACP negotiation. Passive mode: device responds to LACP but does not initiate. Both ends can be active. Configure on ae0: set interfaces ae0 aggregated-ether-options lacp active (initiate LACP). Or passive: set interfaces ae0 aggregated-ether-options lacp passive. Verify: show lacp interfaces (shows negotiation state, partner info, member links). LACP is preferred over static LAG — auto-detects failures.

Q89 What is the Junos interface 'link-mode' setting?

Ans Link mode controls how the interface establishes its connection: full-duplex (simultaneous TX/RX — normal for switches), half-duplex (one direction at a time — older Ethernet). For modern networks, always use full-duplex or auto-negotiation. Setting: set interfaces ge-0/0/0 link-mode full-duplex. Mismatched modes cause: high collision counters, extremely slow throughput, CRC errors. Verify current mode: show interfaces ge-0/0/0 detail | match Duplex.

Q90 What is interface flapping and how do you diagnose it?

Ans Interface flapping = interface repeatedly going up then down, rapidly. Causes: faulty cable (most common), bad SFP/transceiver, duplex negotiation failure, spanning tree issue on switch, faulty NIC on connected device. Diagnose: show log messages | match ge-0/0/0 (see how often 'link up'/'link down' messages appear). show interfaces ge-0/0/0 detail (check carrier-transitions counter — high number indicates flapping). Fix: replace cable, clean SFP, check connected device. Apply hold-time to reduce routing impact.

SECTION 04

Routing Fundamentals

Q91 – Q120

Routing Table · Static Routes · OSPF · BGP · Policy · Default Route

Q91 What is the Routing Information Base (RIB) in Junos?

Ans The RIB (Routing Information Base) is Junos's master routing table — also called the routing table. It contains all routes the device knows from: directly connected networks, static routes, and dynamic routing protocols (OSPF, BGP, ISIS). Each route has: destination prefix, protocol, preference (lower is better), metric, next-hop IP, and interface. View: show route. The best route per destination is installed in the Forwarding Information Base (FIB) and used by the PFE.

TIP RIB = where all routes are collected. FIB = what is actually used for forwarding. Know this difference!

Q92 What is the Forwarding Information Base (FIB) in Junos?

Ans The FIB (also called forwarding table) contains only the BEST route for each destination — the route actually used to forward packets. It is derived from the RIB and downloaded to the PFE hardware. View: show route forwarding-table. The FIB is what the hardware uses — it is optimised for fast lookup. When multiple protocols know a route, RIB selects the best one (using preference/AD) and puts only that in the FIB.

Q93 What is route preference in Junos? (like Administrative Distance)

Ans Route preference is Junos's equivalent of Cisco's Administrative Distance — a value that determines which protocol's route is preferred when multiple protocols know the same destination. Lower preference = more trusted = preferred. Values: Direct (connected) = 0, Static = 5, OSPF Internal = 10, IS-IS L1 = 15, IS-IS L2 = 18, OSPF External = 150, BGP = 170, RIP = 100. Unlike Cisco's AD, Junos preference values are different — do not confuse them!

TIP Junos route preference values are DIFFERENT from Cisco AD values. Learn them separately!

Q94 How do you configure a static route in Junos?

Ans Basic static route: set routing-options static route 192.168.2.0/24 next-hop 10.0.0.2. Default route: set routing-options static route 0.0.0.0/0 next-hop 203.0.113.1. Static route with description: set routing-options static route 192.168.2.0/24 next-hop 10.0.0.2. Discard route (black hole): set routing-options static route 192.168.2.0/24 discard. Verify: show route 192.168.2.0. Static routes have preference 5 in Junos — very trusted (lower than most dynamic protocols).

Q95 What is OSPF in Junos and how does it work?

Ans OSPF (Open Shortest Path First) is a dynamic routing protocol where routers share topology information with each other and calculate the shortest path to every destination. It is 'link-state' – each router builds a complete map of the network. In Junos: configure with 'protocols ospf area [area-id] interface [interface.unit]'. OSPF uses cost as metric (lower = preferred). All routers in an area must share the same topology map for routing to work correctly.

Q96 How do you configure OSPF in Junos?

Ans Minimal OSPF configuration: set protocols ospf area 0 interface ge-0/0/0.0 (enable OSPF on interface, place in area 0). Enable on loopback: set protocols ospf area 0 interface lo0.0 passive (advertise loopback but do not form neighbours on it). Optional Router ID: set routing-options router-id 10.0.0.1. Verify: show ospf neighbor (check adjacency state – should be Full). show ospf route (OSPF routes). show ospf database (link-state database – topology map).

Q97 What is an OSPF area in Junos?

Ans OSPF areas are logical groupings of routers. Area 0 is the backbone – every other area must connect to area 0. Within an area, routers share complete topology information. Between areas, only summary routes are shared (less information = smaller routing tables = better performance). Junos supports: Normal area, Stub area (no external routes), Totally Stub area (no external or inter-area routes), NSSA (Not So Stubby Area – allows some external routes in stub). Configure area type: set protocols ospf area 1 stub.

Q98 What is the OSPF DR and BDR election in Junos?

Ans On multi-access networks (Ethernet with multiple OSPF routers), one router is elected Designated Router (DR) and one Backup DR (BDR). All other routers form OSPF adjacency only with DR/BDR – reduces flooding. Election: highest OSPF priority wins (default 128 in Junos). Tie: highest Router ID wins. Configure priority: set protocols ospf area 0 interface ge-0/0/0.0 priority 200 (high priority = likely DR). Verify: show ospf neighbor (check 'DR' column).

Q99 What is BGP in Junos?

Ans BGP (Border Gateway Protocol) is the routing protocol of the internet – routes traffic between different organisations (Autonomous Systems). In Junos, BGP is used for: internet connectivity (eBGP with ISP), large enterprise routing, MPLS VPNs (iBGP internally), and data centre spine-leaf routing. Junos has excellent BGP support – it is used by most major ISPs running Juniper equipment. BGP uses TCP port 179 and path attributes (AS-path, MED, Local Preference) to select best routes.

Q100 How do you configure BGP in Junos?

Ans Minimal eBGP configuration: set routing-options autonomous-system 65001 (set your AS number). set protocols bgp group EBGP type external (eBGP group). set protocols bgp group EBGP peer-as 65002 (ISP's AS number). set protocols bgp group EBGP neighbor 203.0.113.1 (ISP's BGP peer IP). Verify: show bgp summary (see all BGP peers, state should be 'Established', shows prefixes received). show bgp neighbor 203.0.113.1 detail (detailed peer info).

Q101 What is routing policy in Junos?

Ans Routing policies control which routes are accepted from other protocols and which routes are advertised to neighbours. Think of policy as a filter on route information. Two types: import policy (what routes to accept from neighbours), export policy (what routes to advertise to neighbours). Policies consist of terms, each with: match conditions (from clauses) and actions (then clauses – accept, reject, or modify attributes). Policies are applied to routing protocols.

TIP *Routing policy is a core Junos concept – every production network uses policies!*

Q102 How do you create a basic routing policy in Junos?

Ans Create policy: set policy-options policy-statement ACCEPT-DEFAULTS term 1 from route-filter 0.0.0.0/0 exact (match default route). set policy-options policy-statement ACCEPT-DEFAULTS term 1 then accept (accept it). set policy-options policy-statement ACCEPT-DEFAULTS term 2 then reject (reject everything else). Apply to BGP import: set protocols bgp group EBGP import ACCEPT-DEFAULTS. Without explicitly applying, policy does nothing. Always end policies with explicit accept or reject.

Q103 What is a prefix-list in Junos?

Ans A prefix-list is a named list of IP prefixes used in routing policies to match specific networks. Create: `set policy-options prefix-list MY-NETWORKS 192.168.1.0/24`, `set policy-options prefix-list MY-NETWORKS 10.0.0.0/8`. Use in policy: `set policy-options policy-statement MY-POLICY term 1 from prefix-list MY-NETWORKS`. Cleaner than listing individual route-filters in every policy — define once, reuse everywhere. Verify: `show policy-options prefix-list MY-NETWORKS`.

Q104 What is a community in BGP (Junos)?

Ans A BGP community is a tag attached to a route that carries additional information. Format: `AS:value` (e.g. `65001:100`). Uses: tell neighbour how to handle the route, group routes for policy matching, signal route purpose. Junos well-known communities: `no-export` (do not advertise outside AS), `no-advertise` (do not advertise to any neighbour). Set community: `set policy-options policy-statement ADD-COMMUNITY term 1 then community add MY-COMM`. Define community: `set policy-options community MY-COMM members 65001:100`.

Q105 What is the local-preference attribute in BGP?

Ans Local preference is used within an AS (iBGP) to prefer one exit point over another. Higher local preference = preferred path. Default: 100. If router A receives the same route from two different iBGP peers, the one with higher local-preference wins. Modify with policy: `set policy-options policy-statement SET-LOCALPREF term 1 from neighbor 10.0.0.1 then local-preference 200`. Only propagated within your AS (not sent to eBGP peers). Used to control which ISP is preferred for outbound traffic.

Q106 What is MED (Multi-Exit Discriminator) in BGP?

Ans MED suggests to a neighbouring AS which path to use when entering your AS. Lower MED = preferred. Only compared between routes from same AS. Set MED: `set policy-options policy-statement SET-MED term 1 then metric 100`. Used to influence inbound traffic — you tell your ISP 'please use this link to reach us (lower MED), use the other link only as backup (higher MED)'. Only works if ISP honours MED values — not guaranteed.

Q107 What is AS-path in BGP?

Ans AS-path is the list of Autonomous System numbers a BGP route has passed through to reach the current router. Example: 65002 65010 65100 means the route passed through three AS numbers. BGP prefers routes with shorter AS-paths (fewer hops through different organisations). AS-path prepending: artificially add your own AS number multiple times to make a route look less preferred – useful for traffic engineering (make ISP B path look longer so traffic uses ISP A).

Q108 What is a routing instance in Junos?

Ans A routing instance is a separate, independent routing and forwarding table on a single device. Like having multiple virtual routers. Uses: VRF (VPN Routing and Forwarding) for MPLS L3VPNs, management separation (keep management traffic separate from data), testing. Each routing instance has its own: routing table, interfaces, and routing protocols. Default: 'inet.0' routing table. Create: set routing-instances VPN-A instance-type vrf. View instance routes: show route table VPN-A.

Q109 What is ECMP (Equal-Cost Multi-Path) in Junos?

Ans ECMP allows a router to forward traffic across multiple paths that have equal cost to the same destination – load balancing across multiple links. Enable: set routing-options forwarding-table export ECMP-POLICY and in the policy set then load-balance per-packet or per-flow. Per-flow (default): all packets of a TCP session take the same path (better for TCP). Per-packet: round-robin across paths (may cause reordering). Verify: show route forwarding-table destination [prefix] (see multiple next-hops).

Q110 What is OSPF passive interface in Junos?

Ans A passive interface runs OSPF (so the connected network is advertised) but does NOT form OSPF neighbour relationships or send/receive hello packets. Use on interfaces connected to hosts (PCs, printers) – they do not run OSPF. Configure: set protocols ospf area 0 interface ge-0/0/3.0 passive. Benefits: hosts do not receive OSPF hello packets (less traffic), security (attackers cannot inject fake OSPF routes from user ports), cleaner network. Alternatively: configure just lo0 as passive – loopback should always be passive.

Q111 What is redistribute in Junos routing?

Ans Redistribution imports routes from one routing protocol into another using export policies. Example: export OSPF routes into BGP. Create policy: set policy-options policy-statement OSPF-INTO-BGP term 1 from protocol ospf (match OSPF routes) then accept. Apply to BGP export: set protocols bgp group IBGP export OSPF-INTO-BGP. Junos redistribution is clean and explicit — you must write a policy and apply it. No automatic redistribution like some other platforms. This gives precise control.

Q112 What is the Junos route filter in policies?

Ans Route filters match specific IP prefixes in routing policies. Options: exact — match exactly that prefix (192.168.1.0/24 exact = only /24). orlonger — match prefix and any more specific (192.168.1.0/24 orlonger = /24, /25, /26, ...). upto — match prefix up to specified length (192.168.0.0/16 upto /24 = /16 through /24). longer — more specific only (not the prefix itself). Example: set policy-options policy-statement FILTER term 1 from route-filter 10.0.0.0/8 orlonger.

Q113 What is 'show route detail' in Junos?

Ans show route [prefix] detail shows all attributes of a route: which protocol learned it, next-hop IP and interface, metric, preference, route age, AS-path (for BGP), communities, local-preference, MED. For BGP routes: shows all received paths (active path marked with *). Essential for troubleshooting routing: 'why is traffic going through Router B instead of Router A?' — check show route detail to see route attributes and why Router B's path won.

Q114 What is OSPF authentication in Junos?

Ans OSPF authentication ensures only trusted routers can become neighbours. Types: Simple (MD5 or SHA are preferred; plain text is insecure and should not be used in production). Configure MD5: set protocols ospf area 0 interface ge-0/0/0.0 authentication md5 1 key MySecretKey. Both ends must use the same key ID and key. Benefit: prevents a rogue device from connecting and injecting false routing information. In production, always use OSPF authentication on all interfaces.

Q115 What is route damping in BGP (Junos)?

Ans BGP route damping suppresses unstable routes (routes that keep being withdrawn and re-advertised) to protect the routing table. Each time a route flaps (goes away and comes back), it accumulates penalty. When penalty exceeds threshold, route is suppressed (not used). Penalty decays over time. Configure: `set protocols bgp damping`. Caution: route damping can delay convergence after legitimate route changes. Junos implements it but many networks disable it (RIPE recommends against it for most cases).

Q116 What is the difference between iBGP and eBGP?

Ans eBGP (external BGP): BGP session between routers in DIFFERENT Autonomous Systems. Typical use: your router connects to ISP's router. TTL = 1 (must be directly connected by default). iBGP (internal BGP): BGP session between routers in the SAME Autonomous System. Typical use: distribute BGP routes internally. TTL = 255 (can be across multiple hops – use 'multihop' option). Key iBGP rule: routes learned via iBGP are NOT re-advertised to other iBGP peers – requires full mesh or Route Reflector.

Q117 What is a route reflector in BGP?

Ans In iBGP, every router must peer with every other router (full mesh) – becomes unscalable with many routers. A Route Reflector (RR) is a central BGP peer that re-advertises iBGP routes to its clients – clients only need to peer with the RR, not with each other. Configure Junos as RR: `set protocols bgp group IBGP cluster 10.0.0.1`. Clients peer with RR as normal iBGP. RR adds cluster-id and originator-id attributes to prevent routing loops. Dramatically reduces iBGP peer count.

Q118 How do you troubleshoot BGP in Junos?

Ans BGP troubleshooting commands: `show bgp summary` (all peers: state Active=trying to connect, Idle=waiting, Established=working). `show bgp neighbor [IP]` (detailed peer info – open/keepalive timers, error messages). `show bgp neighbor [IP] received-routes` (all routes received from peer). `show bgp neighbor [IP] advertised-routes` (what we are sending to peer). `show route protocol bgp` (BGP routes in routing table). Common issues: wrong AS number, wrong neighbor IP, firewall blocking TCP 179, authentication mismatch.

Q119 What is the routing policy 'then' action in Junos?

Ans The 'then' clause in a routing policy term specifies what to do with matched routes: accept – route passes the filter. reject – route is dropped/not used. next-term – continue to next term without accepting/rejecting. next policy – try the next applied policy. Modify attributes (and still accept): then local-preference 200, then metric 10, then community add MY-COMM, then as-path-prepend 65001. If no 'then' action is specified and conditions match, Junos evaluates the next term.

Q120 What is 'show route forwarding-table' and when do you use it?

Ans show route forwarding-table shows the actual FIB – what the PFE uses to forward packets. This may differ from show route (RIB) because: multiple protocols may know a route but only the best goes to FIB. show route forwarding-table destination 192.168.1.0/24 – check if a specific route is in the FIB. Use when: ping fails but show route shows correct route – check if it made it to the FIB. Also shows: multicast routes, MPLS labels, and load-balancing entries.

SECTION 05

Switching & VLANs

Q121 –
Q150

EX Series · VLANs · STP · IRB · Trunking · Port Security · Layer 3 Switching

Q121 What is the Junos EX series switch?

Ans Junos EX series are Juniper's enterprise Ethernet switches. Models: EX2300 (access layer, PoE options), EX3400 (access/distribution, high port density), EX4300 (stackable, 10GbE uplinks), EX4650 (25/100GbE data centre), EX9200 (modular core switch). All run full Junos OS – same CLI commands as routers. Key advantage: if you know Junos routing from a router, switching configuration uses the same familiar CLI syntax and tools.

TIP EX switches use the same Junos CLI as routers – one skill set for all Juniper devices!

Q122 How do VLANs work in Junos?

Ans VLANs (Virtual LANs) separate devices into logical broadcast groups on the same physical switch. In Junos: define VLAN: set vlans SALES vlan-id 100. Assign to access port: set interfaces ge-0/0/1 unit 0 family ethernet-switching interface-mode access vlan members SALES. For multiple VLANs on a port (trunk): set interfaces ge-0/0/0 unit 0 family ethernet-switching interface-mode trunk vlan members [VLAN-list or 'all']. Verify: show vlans (list all VLANs), show ethernet-switching interface (port VLAN assignments).

Q123 What is the difference between access mode and trunk mode in Junos?

Ans Access mode: port belongs to one VLAN only. Frames are untagged (no VLAN tag). Used for: connecting PCs, printers, IP phones to a single VLAN. Trunk mode: port carries multiple VLANs with 802.1Q tags. Used for: uplinks between switches, connections to routers (ROAS), hypervisor uplinks (carries all VM VLANs). In Junos: interface-mode access vs interface-mode trunk under family ethernet-switching. Default: Junos EX ports are configured as access ports in the default VLAN.

Q124 What is the native VLAN equivalent in Junos?

Ans In Junos, the equivalent of Cisco's native VLAN is the 'native-vlan-id' on a trunk interface. Traffic in this VLAN is sent untagged on the trunk link. Configure: set interfaces ge-0/0/0 unit 0 family ethernet-switching interface-mode trunk native-vlan-id 1. Both ends of the trunk must agree on native VLAN – mismatch causes connectivity problems. Security best practice: change native VLAN from the default VLAN 1 to an unused VLAN to prevent VLAN hopping attacks.

Q125 What is Spanning Tree Protocol in Junos?

Ans STP (IEEE 802.1D) or RSTP (IEEE 802.1w) prevents Layer 2 loops in redundant switched networks. Junos EX switches run RSTP by default (faster than original STP – converges in 1-2 seconds). Configure: set protocols rstp. STP elects a Root Bridge (switch with lowest Bridge ID). Non-root switches calculate their best path to root and block redundant paths. Verify: show spanning-tree bridge (see root bridge info), show spanning-tree interface (see port roles).

Q126 How do you set the STP root bridge priority in Junos?

Ans Lower priority = more likely to become Root Bridge. set protocols rstp bridge-priority 4096 (lower than default 32768 – becomes Root Bridge). Or use the convenient option: set protocols rstp interface ge-0/0/0 edge (PortFast equivalent for edge ports). For VSTP (per-VLAN STP): set protocols vstp vlan 100 bridge-priority 4096. Verify current root bridge: show spanning-tree bridge. Best practice: manually set the most central, reliable switch as Root Bridge – do not leave election to chance.

Q127 What is VSTP in Junos?

Ans VSTP (VLAN Spanning Tree Protocol) is Juniper's version of Cisco's PVST+ – runs a separate STP instance per VLAN. This allows different VLANs to have different root bridges, enabling load balancing across redundant uplinks. Configure: set protocols vstp vlan 100 (enables VSTP for VLAN 100). set protocols vstp vlan 100 bridge-priority 4096 (make this switch root for VLAN 100). Verify: show spanning-tree interface (check per-VLAN port states). VSTP and RSTP cannot run simultaneously on the same switch.

Q128 What is IRB interface in Junos switching?

Ans IRB (Integrated Routing and Bridging) provides a Layer 3 interface for a VLAN on a Junos EX switch – enabling routing between VLANs without a separate router. Configure IRB for VLAN 100: set interfaces irb unit 100 family inet address 192.168.100.1/24. Link IRB to VLAN: set vlans SALES l3-interface irb.100. Enable routing (on L3 switch): set routing-options... (may need ip routing equivalent). Clients in VLAN 100 use 192.168.100.1 as default gateway.

Q129 What is port security in Junos EX?

Ans Port security limits which MAC addresses can communicate on a switch port. Configure MAC limit: set interfaces ge-0/0/1 unit 0 family ethernet-switching mac-limit 2 action drop (allow max 2 MACs, drop if exceeded). Or use MAC-based authentication (802.1X). For dynamic MAC security: set ethernet-switching-options mac-limit interface ge-0/0/1 action drop. Persistent MAC: configure specific allowed MACs per port. Prevents MAC flooding attacks and unauthorised device connections.

Q130 What is 802.1X authentication in Junos?

Ans 802.1X controls network access at the switch port – requires devices to authenticate before getting network access. Components: Supplicant (end device), Authenticator (Junos switch), Authentication Server (RADIUS). Configure: set protocols dot1x authenticator interface ge-0/0/1 supplicant multiple. set protocols dot1x authenticator authentication-profile-name MY-RADIUS. If authentication fails: port goes to guest VLAN or stays blocked. Verify: show dot1x interface (authentication state per port).

Q131 How do you configure LACP link aggregation on EX switches?

Ans Configure LAG on EX switch: set chassis aggregated-devices ethernet device-count 1 (reserve ae interfaces). set interfaces ge-0/0/0 ether-options 802.3ad ae0 (add to LAG). set interfaces ge-0/0/1 ether-options 802.3ad ae0 (add second member). set interfaces ae0 aggregated-ether-options lacp active (enable LACP). set interfaces ae0 unit 0 family ethernet-switching interface-mode trunk vlan members all. Verify: show lacp interfaces (see negotiation state), show interfaces ae0 (see bundle status).

Q132 What is LLDP-MED in Junos and how does it help IP phones?

Ans LLDP-MED (Link Layer Discovery Protocol – Media Endpoint Discovery) extends LLDP for VoIP phones. It allows a switch to automatically: discover VoIP phones, assign them to the correct voice VLAN, configure QoS settings. The phone sends LLDP-MED messages; the switch responds with voice VLAN configuration. Enable: set protocols lldp-med interface all. No manual voice VLAN configuration needed per port – plug in phone, it auto-configures. Very useful in large office deployments.

Q133 What is IGMP snooping in Junos?

Ans IGMP snooping allows a switch to intelligently forward multicast traffic only to ports that have requested it – instead of flooding multicast to all ports. The switch 'snoops' on IGMP membership messages. Result: only clients watching a multicast stream (e.g. video stream) receive it, reducing unnecessary traffic on all other ports. Enable: `set vlans SALES protocols igmp-snooping`. Verify: `show igmp snooping membership` (see which ports have joined which multicast groups).

Q134 What is storm control in Junos?

Ans Storm control limits broadcast, multicast, or unknown unicast traffic to prevent network flooding. Set bandwidth percentage: `set forwarding-options storm-control interface ge-0/0/1 bandwidth-percentage 10` (block if broadcast exceeds 10% of port bandwidth). Action options: `drop` (silently discard excess), `shutdown` (disable port). Apply: `set interfaces ge-0/0/1 unit 0 family ethernet-switching storm-control DEFAULT`. Protects against broadcast storms from loops, misconfigured devices, or network attacks.

Q135 What is Virtual Chassis in Junos EX?

Ans Virtual Chassis connects multiple Juniper EX switches (up to 10 in some models) and makes them appear as ONE logical switch. All switches share a single management point, single configuration, single spanning tree. Benefits: expand switch ports without adding management complexity, redundancy (if one switch fails, others continue), simplified cabling. Connect switches with dedicated VCPs (Virtual Chassis Ports) or use uplink ports. Command: `show virtual-chassis` (see member status).

Q136 How do you configure a VLAN on Junos EX and verify it?

Ans Complete VLAN setup: Create VLAN: `set vlans HR vlan-id 20`. Assign ports: `set interfaces ge-0/0/5 unit 0 family ethernet-switching interface-mode access vlan members HR`. Configure gateway: `set interfaces irb unit 20 family inet address 192.168.20.1/24`. Link to VLAN: `set vlans HR l3-interface irb.20`. Commit. Verify: `show vlans HR` (see VLAN members and IRB interface), `show interfaces irb.20` (see IP and status), ping 192.168.20.1 from a client in VLAN 20 to test gateway.

Q137 What is the Junos forwarding table for switching?

Ans show ethernet-switching table — displays the MAC address table (CAM table) showing: MAC address, VLAN, interface learned on, and age. The switch uses this table to forward frames to the correct port. If destination MAC is not in table, switch floods the frame to all ports in that VLAN. show ethernet-switching table count — how many entries. clear ethernet-switching table — flush all dynamic entries (they will be relearned). Verify connectivity with this table: 'do I see the destination MAC?'

Q138 What is MSTP (Multiple Spanning Tree Protocol) in Junos?

Ans MSTP (IEEE 802.1s) allows multiple VLANs to be mapped to a smaller number of STP instances — combining the efficiency of VSTP (per-VLAN STP) with less overhead. Instead of one STP instance per VLAN (which can be 1000 instances for 1000 VLANs!), MSTP groups VLANs into regions with shared instances. Configure: set protocols mstp region-name MY-REGION revision 1 vlan 1-100 bridge-instance 1. Useful in large enterprise networks with many VLANs.

Q139 What is the default VLAN in Junos and how does it work?

Ans The default VLAN in Junos (VLAN ID 1) is the VLAN that all ports belong to before being explicitly assigned to another VLAN. Unlike Cisco where VLAN 1 is the native VLAN on trunks, Junos handles this differently. Best practice: do not use VLAN 1 for real traffic — create named VLANs for all networks. Disable unused ports: set interfaces ge-0/0/10 disable. Change the default VLAN behaviour by explicitly assigning every port to a specific named VLAN.

Q140 How do you troubleshoot a VLAN connectivity issue in Junos?

Ans VLAN troubleshooting checklist: 1) Does VLAN exist on switch? (show vlans). 2) Is the access port in the correct VLAN? (show interfaces ge-0/0/x). 3) Is the trunk carrying the VLAN? (show interfaces trunk-port — check vlan members). 4) Does the IRB interface exist and is it up? (show interfaces irb.20). 5) Is IP routing enabled on L3 switch? 6) Can the PC ping the IRB gateway? If not — check PC's default gateway setting. 7) ARP working? (show arp).

Q141 What is the MAC address learning process on a Junos switch?

Ans When a frame arrives on a switch port, Junos records the source MAC address and port in the MAC table (CAM table) – this is learning. Next time a frame is destined for that MAC, the switch sends it directly to the correct port – this is forwarding. If MAC is unknown: switch floods to all ports in that VLAN. MAC entries expire after 300 seconds of inactivity (age-out time). Configure aging: `set ethernet-switching-options aging-time 600`. Disable learning on specific port: use port security features.

Q142 What is PoE (Power over Ethernet) on Junos EX?

Ans Junos EX switches support PoE (IEEE 802.3af: 15W, 802.3at PoE+: 30W, 802.3bt PoE++: 60-100W) – delivers power over Ethernet cables to IP phones, WAPs, cameras. No separate power adapter needed. Manage PoE in Junos: `set poe interface ge-0/0/1 maximum 15400` (set max watts). `show poe interface` – see power consumption per port, total PoE budget used. `set poe interface ge-0/0/1 disable` – cut power to specific port. PoE power budget is shared across all ports – plan carefully.

Q143 What is QinQ (double tagging) in Junos?

Ans QinQ (IEEE 802.1ad) adds TWO 802.1Q VLAN tags to a frame – an outer tag (service provider VLAN) and inner tag (customer VLAN). Used by: service providers to carry multiple customer VLANs across their network without VLAN ID conflicts. Customer has VLANs 1-4094; provider wraps them in their own VLAN. Configure: `set interfaces ge-0/0/0 flexible-vlan-tagging`, then set inner and outer VLAN IDs in the unit configuration. Allows $4094 \times 4094 = 16$ million unique VLAN combinations.

Q144 What is the Junos 'show ethernet-switching statistics' command?

Ans `show ethernet-switching statistics` displays per-VLAN and per-interface switching statistics: total frames forwarded, flooded, filtered, and errors. Useful for: identifying VLANs with excessive flooding (indicates missing MAC entries or misconfigurations), measuring per-port traffic rates, detecting broadcast storms (very high flood count). Supplement with: `show interfaces ge-0/0/x statistics` for physical-level counters like input/output errors.

Q145 What is a private VLAN (PVLAN) in Junos?

Ans Private VLANs (PVLANs) allow ports within the same VLAN to be isolated from each other while still sharing the same IP subnet and gateway. Use case: hosting environment where multiple servers share a subnet but must not communicate directly (security isolation). Types: promiscuous port (can talk to all), isolated port (can only talk to promiscuous port), community port (can talk within community and to promiscuous). Configure under family ethernet-switching with private-vlan settings.

Q146 What is rapid per-VLAN spanning tree (RPVST+) in Junos?

Ans RPVST+ combines Rapid STP convergence (1-2 seconds) with per-VLAN STP instances — best of both worlds. Junos calls this VSTP + RSTP enabled together. Each VLAN has its own STP topology but uses RSTP for fast failover. Useful for: load balancing traffic across redundant uplinks (different VLANs use different uplinks), fast recovery when a link fails (RSTP convergence). Configure: set protocols vstp (enables per-VLAN RSTP). Show: show spanning-tree bridge instance [vlan-id].

Q147 What are the STP port roles in Junos?

Ans STP assigns roles to each port: Root Port: the port closest to Root Bridge on each non-root switch (one per switch). Designated Port: the port that best connects a network segment to the Root Bridge (one per segment — forwards traffic). Alternate Port: blocked port that provides an alternative path to the root — instantly unblocked if root port fails (RSTP improvement over STP). Backup Port: blocked port providing redundancy for a designated port on the same segment. Verify: show spanning-tree interface.

Q148 How does RSTP edge port work in Junos?

Ans RSTP edge port (equivalent to Cisco PortFast) allows a port connected to an end device to immediately transition to forwarding without going through STP listening and learning states. Configure: set protocols rstp interface ge-0/0/5 edge. The port skips the 30-50 second STP wait. If the edge port receives a BPDU (from a switch), it automatically loses edge status and goes through normal STP convergence — this is the built-in safety mechanism. Always configure on ports connecting to PCs, printers, and IP phones.

Q149 What is the Junos EX management VLAN?

Ans The management VLAN carries management traffic (SSH, SNMP, Syslog) to the switch. Best practice: use a separate dedicated management VLAN (e.g. VLAN 99) not used for regular user traffic. Configure: set vlans MGMT vlan-id 99. Create management IRB: set interfaces irb unit 99 family inet address 10.99.1.1/24. Assign management port to this VLAN. Restrict SSH access to management IP only. This isolates management traffic from user traffic and adds security.

Q150 What is Junos Enhanced Layer 2 Software (ELS) and why does it matter?

Ans ELS (Enhanced Layer 2 Software) is an updated switching configuration format introduced in newer Junos versions for EX and QFX switches. ELS changes: uses 'vlans VLAN-NAME' (not 'vlan-id' under interfaces), uses 'interface-mode access/trunk', uses 'irb' interfaces. Non-ELS (older): used 'family ethernet-switching' under unit 0. Some EX models use ELS, others do not – check with 'show version' and Juniper documentation. JNCIA questions may specify which model/syntax to use.

SECTION 06

Security Fundamentals (SRX)

Q151 –
Q180

Firewall Filters · Zones · Security Policies · NAT · IPS · VPN · Hardening

Q151 What is the SRX series in Junos?

Ans Juniper SRX series are security gateways (firewalls) running full Junos OS. Models: SRX300/320/340/345/380 (branch office), SRX1500/3600 (mid-range), SRX5400/5600/5800 (high-end carrier). All share the same Junos CLI — if you know SRX300 config, you know SRX5800 conceptually. SRX is a stateful firewall, VPN gateway, IPS (Intrusion Prevention), NAT device, and router in one box. Used by enterprises and service providers for internet edge security.

TIP SRX is one of the most widely deployed firewalls in enterprises globally. Junos makes it accessible!

Q152 What is a security zone in Junos SRX?

Ans A security zone is a logical grouping of interfaces that share the same security requirements and trust level. Common zones: trust (internal LAN — trusted network), untrust (internet — not trusted), dmz (servers accessible from internet — semi-trusted). Interfaces are assigned to zones: set security zones security-zone trust interfaces ge-0/0/1.0. Traffic between zones is controlled by security policies. Traffic within the same zone is allowed by default.

Q153 What is a security policy in Junos SRX?

Ans A security policy defines what traffic is allowed between zones. Like a firewall rule. Configure: set security policies from-zone trust to-zone untrust policy ALLOW-WEB match source-address any destination-address any application junos-http. set security policies from-zone trust to-zone untrust policy ALLOW-WEB then permit. Apply: policies are automatically matched top-to-bottom for zone pair traffic. Without a matching policy: traffic is denied (implicit deny all). Verify: show security policies (list all policies).

Q154 What is a firewall filter in Junos (stateless)?

Ans A firewall filter is a stateless packet filter — like an ACL on a Cisco router. It examines each packet independently (does not track connections). Use for: protecting the router's own control plane, rate limiting specific traffic, marking traffic for QoS. Configure: set firewall filter PROTECT-RE term ALLOW-SSH from source-address 192.168.1.0/24 protocol tcp destination-port 22. set firewall filter PROTECT-RE term ALLOW-SSH then accept. Apply to interface: set interfaces ge-0/0/0 unit 0 family inet filter input PROTECT-RE.

Q155 What is the difference between a firewall filter and a security policy in Junos?

Ans Firewall filter (stateless): examines each packet independently, does not track TCP sessions, used on routers for ACL-like filtering, applied per-interface per direction, uses 'then accept/reject/discard'. Security policy (stateful): tracks complete TCP/UDP sessions, knows if packet is part of established session, used on SRX firewalls between zones, automatically allows return traffic for permitted sessions. Use firewall filters on routers, security policies on SRX firewalls.

Q156 What is NAT in Junos SRX?

Ans NAT (Network Address Translation) on SRX translates IP addresses. Types: Source NAT (SNAT): translates source IP — used for internet access from private network. Destination NAT (DNAT): translates destination IP — used for port forwarding (publish a server behind SRX). Static NAT: one-to-one mapping. Configure source NAT pool: set security nat source pool INET-POOL address 203.0.113.1/32. Configure rule: set security nat source rule-set OUTBOUND rule SNAT-RULE match source-address 192.168.1.0/24. set security nat source rule-set OUTBOUND rule SNAT-RULE then source-nat pool INET-POOL.

Q157 What is application in Junos security policy?

Ans In Junos security policies, 'application' defines which protocol/port the policy matches. Junos has hundreds of predefined applications: junos-http (HTTP, port 80), junos-https (HTTPS, port 443), junos-ssh (SSH, port 22), junos-dns (DNS, port 53), junos-ftp (FTP). Create custom: set applications application MY-APP protocol tcp destination-port 8080. Use in policy: set security policies from-zone trust to-zone untrust policy MY-POLICY match application MY-APP. Predefined apps handle port ranges and multi-port applications cleanly.

Q158 What is AppSecure in Junos SRX?

Ans AppSecure is Juniper's application identification and security system on SRX. It identifies applications regardless of port — useful when applications use non-standard ports (e.g. Skype on random ports). Components: Application Identification (AppID) — identifies applications by behaviour. AppTrack — logs application usage. AppFW — creates policies based on application name. AppQoS — applies QoS based on application. Requires UTM license. Enable: set services application-identification.

Q159 What is IPS/IDS on Junos SRX?

Ans IPS (Intrusion Prevention System) inspects traffic for known attack signatures and blocks malicious traffic. IDS (Intrusion Detection System) just alerts without blocking. SRX IPS: uses Juniper's signature database (downloaded from Juniper). Configure: set security idp idp-policy MY-IDP rulebase-ips rule 1 match attacks predefined-attack-groups 'Critical'. set security idp idp-policy MY-IDP rulebase-ips rule 1 then action drop-packet. Apply to security policy: set security policies... then permit application-services idp-policy MY-IDP. Update signatures regularly.

Q160 What is UTM (Unified Threat Management) in Junos?

Ans UTM on SRX combines multiple security functions: Antivirus (scans files in HTTP/FTP/SMTP/POP3), Anti-spam (blocks spam email), Content Filtering (blocks websites by category), Web Filtering (Websense/Juniper category lists). Requires UTM license. Configure feature profile: set security utm feature-profile web-filtering juniper-enhanced profile MY-PROFILE. Apply in security policy: set security policies... then permit application-services utm-policy MY-UTM. Very useful for branch offices needing comprehensive security in one box.

Q161 What is VPN in Junos SRX?

Ans Junos SRX supports multiple VPN types: IPsec Site-to-Site VPN: encrypts traffic between two sites over internet. Route-based VPN (preferred): uses st0 tunnel interface and routing to direct traffic. Policy-based VPN (simpler): uses security policies to define interesting traffic. SSL/TLS VPN: Junos Pulse or SA series for remote user VPN. Dynamic VPN (DynVPN): allows remote users to connect via IPsec without dedicated hardware. IKEv2 is the modern standard.

Q162 How do you configure a basic IPsec VPN in Junos?

Ans IPsec VPN steps: Define IKE proposal (Phase 1 algorithms): set security ike proposal MY-IKE-PROP authentication-method pre-shared-keys dh-group group14 encryption-algorithm aes-256-cbc. Create IKE policy: set security ike policy MY-IKE-POL pre-shared-key ascii-text MySecret proposals MY-IKE-PROP. Configure IKE gateway: set security ike gateway MY-GW address [peer-IP] ike-policy MY-IKE-POL external-interface ge-0/0/0.0. Define IPsec proposal and VPN. Associate st0 interface. Configure routing. Verify: show security ike security-associations.

Q163 What is screen options in Junos SRX?

Ans Screen options (IDS screens) protect against common network attacks at the SRX. Types: SYN flood protection (limits SYN rate to prevent DoS), ICMP flood protection, UDP flood, IP spoofing detection, port scan detection. Configure: set security screen ids-option MY-SCREEN icmp flood threshold 1000. Apply to zone: set security zones security-zone untrust screen MY-SCREEN. Verify: show security screen statistics (see blocked attacks). Screens are processed BEFORE security policies — first line of defence.

Q164 What is the Junos SRX flow mode vs packet mode?

Ans Flow mode: stateful firewall mode — SRX tracks connection state, understands TCP sessions, allows return traffic automatically. Default for security use. All security policies and NAT apply in flow mode. Packet mode: stateless, like a router — processes each packet independently, no session tracking. Used for: GRE and other tunnel protocols that flow mode cannot handle, high-performance routing with simple ACLs. Most SRX deployments use flow mode. Configure: set security forwarding-options family inet6 mode flow-based.

Q165 What is address book in Junos SRX?

Ans Address books are named IP address objects used in security policies — cleaner than using raw IP addresses in policy rules. Global address book: set security address-book global address WEBSERVER 192.168.1.100/32. Zone-specific address book (older style): set security zones security-zone trust address-book address PC-SUBNET 192.168.1.0/24. Use in policy: set security policies from-zone trust to-zone untrust policy MY-POLICY match source-address PC-SUBNET. Makes policies readable and easier to update — change the address object, all policies using it update automatically.

Q166 What is 'protect-re' (Routing Engine protection) in Junos?

Ans The Routing Engine runs the Junos OS — if it is overwhelmed by traffic (DDoS targeting the router itself), the device becomes unmanageable. Protect-RE uses a firewall filter to limit traffic destined for the RE itself. Best practice: apply a filter on lo0.0 (loopback) to: allow only your management hosts to SSH, allow OSPF from known neighbours, rate limit ICMP, block everything else. set interfaces lo0 unit 0 family inet filter input PROTECT-RE. This is critical for production security.

Q167 What is Junos Security Intelligence?

Ans Security Intelligence (also called ThreatFeed) enables SRX to dynamically block known malicious IP addresses and URLs based on regularly updated threat feeds from Juniper's cloud service. Categories: command-and-control (C&C;) servers, infected hosts, malicious URLs. SRX downloads feed updates automatically. Configure: set security intelligence policy MY-SI-POLICY category CC default-rule then block connection. Apply to security policy. No need to manually maintain blacklists – automatically updated.

Q168 What is transparent mode (Layer 2) in Junos SRX?

Ans In transparent mode, SRX acts as a Layer 2 bridge (like a switch) rather than a router – traffic passes through without IP routing. Useful for: inserting SRX into an existing network without changing IP addressing (no routing changes needed). Configure: set security forwarding-options family inet mode transparent. Configure bridge groups instead of zones. Can still apply security policies, IPS, and UTM in transparent mode. Good for brownfield deployments where changing IP addressing is not feasible.

Q169 What are application services in Junos security policy?

Ans Application services are security features applied within a security policy. Available services: idp-policy (IPS inspection), utm-policy (antivirus, web filtering), ssl-proxy (SSL inspection), application-identification (AppID for Layer 7 visibility). Configure: set security policies from-zone trust to-zone untrust policy WEB-BROWSE then permit application-services idp-policy MY-IPS utm-policy MY-UTM. All selected services apply to matching traffic. Each service requires appropriate license.

Q170 What is a DMZ in Junos SRX?

Ans DMZ (Demilitarized Zone) is a separate security zone for publicly accessible servers. Configure: Create DMZ zone: set security zones security-zone dmz interfaces ge-0/0/2.0. Create security policies: allow internet to reach web server in DMZ on port 80/443. Allow DMZ server to reach internet (for updates) – only specific destinations. Block DMZ from accessing internal trust zone directly. Allow internal trust to manage DMZ servers. This isolates public servers – if web server is compromised, attacker is in DMZ not internal network.

Q171 What is Junos CoS (Class of Service) on SRX/routers?

Ans CoS (Class of Service) is Junos's QoS system – prioritises traffic to ensure important applications (VoIP, video) work well even when network is congested. Components: Traffic classification (identify traffic – by IP, protocol, or DSCP marks), Queuing (4-8 queues with different priorities), Scheduling (how much bandwidth each queue gets), Shaping (limit traffic rate to not overwhelm slow links). Configure: set class-of-service interfaces ge-0/0/0 unit 0 classifiers dscp MY-CLASSIFIER.

Q172 What is RADIUS and TACACS+ in Junos security?

Ans RADIUS and TACACS+ are AAA (Authentication, Authorisation, Accounting) protocols for centralised management access control. Configure RADIUS: set access radius-server 192.168.1.100 secret MySecret. set system authentication-order [radius, password]. set system radius-server 192.168.1.100 port 1812. TACACS+: set system tacplus-server 192.168.1.101 secret MyKey. Benefits: one password change applies to all devices, logs every command, different access levels per user. Essential for enterprise networks with many Juniper devices.

Q173 What is Junos session table on SRX?

Ans The session table records all active firewall connections. Each entry: source/destination IP, protocol, ports, security policy that permitted it, NAT translation. View: show security flow session (all active sessions – may be millions on busy firewall). show security flow session source-prefix 192.168.1.5 (sessions from specific host). Session table size is limited by hardware – high-end SRX handles millions of sessions. Useful for: verifying traffic is being processed, troubleshooting connectivity issues, seeing NAT translations.

Q174 What is NAT hairpin (hairpin NAT) in Junos?

Ans Hairpin NAT (also called NAT loopback or reflection) allows internal clients to reach an internally-hosted server using the server's public IP address – as if coming from the internet. Without it: internal client tries to reach 203.0.113.10 (server's public IP), but traffic never leaves the network. With hairpin NAT: SRX translates the destination to the server's private IP and the source to SRX's internal IP so the server can reply. Configure with destination NAT + source NAT for the internal traffic.

Q175 What is dynamic VPN in Junos?

Ans Dynamic VPN allows remote users to connect via IPsec without client configuration – they use a web browser to download the VPN client automatically. The user navigates to the SRX's external IP in a browser, authenticates, client downloads and connects automatically. Useful for: employee remote access when IT cannot pre-configure every laptop. Configure: set security dynamic-vpn access-profile MY-PROFILE (defines user authentication). set security dynamic-vpn clients MY-CLIENTS ipsec-vpn MY-VPN. Verify: show security dynamic-vpn users.

Q176 What is GRE (Generic Routing Encapsulation) in Junos?

Ans GRE wraps one packet inside another – creates a virtual tunnel. In Junos: configure gr-0/x/x (GRE) tunnel interface. tunnel source [local-IP] tunnel destination [remote-IP]. Assign IP to tunnel interface. Add routes pointing to tunnel. Use case: connect two OSPF routers across an internet path that only supports IP routing (OSPF cannot run natively over internet – run it over GRE tunnel). Note: GRE adds overhead (24 bytes). Set MTU accordingly: tunnel interface MTU = physical MTU - 24.

Q177 What is PKI (Public Key Infrastructure) in Junos?

Ans PKI manages digital certificates for secure communication. Junos uses certificates for: IKEv2 IPsec VPN (certificate-based authentication instead of pre-shared keys – more scalable), SSL management access (HTTPS for J-Web). Commands: request security pki generate-key-pair certificate-id MY-CERT (generate key pair). request security pki local-certificate generate-self-signed certificate-id MY-CERT (self-signed for testing). For production: get certificate signed by CA. Verify: show security pki certificate detail.

Q178 What is ALG (Application Layer Gateway) in Junos SRX?

Ans ALGs handle protocols that embed IP addresses or ports inside their application data – which NAT breaks. Example: FTP embeds port numbers inside the FTP control session data. Without ALG, NAT translates outer IP but misses the embedded port. ALG fixes this by inspecting and modifying application data. Junos SRX has ALGs for: FTP, TFTP, DNS, H.323, SIP, RTSP. Disable specific ALG if causing issues: set security alg ftp disable. Verify: show security alg status.

Q179 What is high availability (HA) cluster on SRX?

Ans Two SRX devices can form a chassis cluster for Active/Passive high availability. Both nodes use same configuration, session tables are synchronised. Control link: heartbeat between nodes. Fabric link: synchronise session state (for stateful failover — existing sessions survive failover). Node 0 is typically Active. If it fails, Node 1 takes over all traffic. Configure cluster: set chassis cluster reth-count 2 redundancy-group 1. Configure redundant Ethernet (reth) interfaces on both nodes. Verify: show chassis cluster status.

Q180 What is traffic logging in Junos SRX?

Ans Traffic logging records what traffic passes through or is blocked by the SRX. Essential for: security incident investigation, compliance, troubleshooting connectivity issues. Types: traffic log (per-session record), event log (system events). Enable session logging: set security policies from-zone trust to-zone untrust policy MY-POLICY then log session-init (log when session starts) session-close (log when session ends). View logs: show log [logfile]. Forward to Syslog server: set system syslog host 192.168.1.200 any any.

SECTION 07

MPLS & Traffic Engineering

Q181 –
Q210

MPLS Labels · LDP · RSVP · L2VPN · L3VPN · MPLS Traffic Engineering · LSP

Q181 What is MPLS and why is it used?

Ans MPLS (Multiprotocol Label Switching) forwards packets using short labels instead of IP routing table lookups – much faster and enables advanced services. How it works: ingress router adds a label (short number) to packets. Core routers forward by label (no IP lookup needed). Egress router removes label. Why use it? Creates guaranteed-bandwidth circuits (like a dedicated wire), enables VPNs (L2/L3), traffic engineering (control exactly which path traffic takes), supports services providers.

TIP *MPLS is fundamental for service provider networking – Junos JNCIA covers basics.*

Q182 What is an LSP (Label Switched Path) in MPLS?

Ans An LSP is the end-to-end virtual circuit through the MPLS network that a labelled packet follows. Think of it like a dedicated highway from city A to city B. Each router in the path is called an LSR (Label Switched Router). The first router (ingress/head-end) adds the label. Middle routers (transit LSRs) swap the label. Last router (egress/tail-end) removes the label. Two protocols set up LSPs: LDP (dynamic, follows IGP path) and RSVP (with explicit path and bandwidth reservation).

Q183 What is LDP (Label Distribution Protocol)?

Ans LDP automatically distributes labels between MPLS-enabled routers, building label mappings for all IP prefixes. It follows the IGP (OSPF/ISIS) shortest path – no manual path configuration needed. Simple to deploy. Enable on Junos: `set protocols ldp interface ge-0/0/0.0`. LDP neighbours are automatically discovered. Verify: `show ldp neighbor` (check adjacency), `show ldp database` (label bindings), `show mpls label-switched-path` (active LSPs). LDP is easier to configure than RSVP but offers no traffic engineering.

Q184 What is RSVP (Resource Reservation Protocol)?

Ans RSVP allows explicit path definition and bandwidth reservation for MPLS LSPs — this is traffic engineering. You can control exactly which routers an LSP traverses, reserve guaranteed bandwidth on each link, set up backup LSPs for fast reroute. Configure RSVP LSP: set protocols mpls label-switched-path MY-LSP to 10.0.0.5 (destination). set protocols mpls label-switched-path MY-LSP bandwidth 100m (reserve 100 Mbps). Verify: show mpls label-switched-path detail (path, bandwidth, utilisation).

Q185 What is an MPLS label and what does it look like?

Ans An MPLS label is a 32-bit value inserted between the Layer 2 header and Layer 3 IP header (called 'shim header'). Fields: Label value (20 bits: 0-1048575), TC/EXP (3 bits: traffic class for QoS), S (1 bit: bottom of stack — 1 = last label), TTL (8 bits: like IP TTL, prevents loops). Labels are locally significant — each router assigns its own labels. Label 0 = IPv4 explicit null, Label 3 = implicit null (pop label before forwarding). Multiple labels can be stacked.

Q186 What is label stacking in MPLS?

Ans Multiple MPLS labels can be stacked on a single packet — each layer of MPLS service adds a label. Example: a VPN packet has an inner VPN label (identifies the customer VPN) and an outer transport label (for routing through the MPLS network). The S (bottom of stack) bit = 1 on the innermost label. Transit LSRs only look at the outermost label. This enables complex services: L3VPN over MPLS, pseudowire (L2VPN), LDP over RSVP tunnels — all using label stacking.

Q187 What is MPLS L3VPN (Layer 3 VPN)?

Ans MPLS L3VPN (RFC 4364) provides IP connectivity between customer sites using MPLS backbone. Customers connect to PE (Provider Edge) routers. PE routers have VRF (Virtual Routing and Forwarding) instances per customer. Customer routes are exchanged via MP-BGP (Multi-Protocol BGP with VPNv4 address family). Traffic flows: customer A → PE1 → MPLS core (labelled) → PE2 → customer A site 2. Customers are completely isolated from each other even though they share the provider backbone.

Q188 What is MPLS L2VPN (Layer 2 VPN)?

Ans MPLS L2VPN extends Layer 2 connectivity over an MPLS backbone. Types: VPWS (Virtual Private Wire Service): point-to-point Layer 2 circuit (pseudowire) — like a leased line. VPLS (Virtual Private LAN Service): multipoint Layer 2 service — like an Ethernet switch distributed across MPLS network. Use cases: connecting customer sites at Layer 2 (transparent to routing), extending VLANs across MPLS network. In Junos: configured under 'protocols l2circuit' (VPWS) or 'protocols vpls' (VPLS).

Q189 What is PE, CE, and P router in MPLS VPN?

Ans PE (Provider Edge): service provider's edge router. Connects to customer CE routers. Maintains VRF tables per customer. Runs MP-BGP to exchange VPN routes with other PEs. Most complex role. P (Provider): core router in MPLS backbone. Only does label switching — no IP routing for customer traffic. Just swaps labels. CE (Customer Edge): customer's router connecting to the provider's PE. Runs normal routing (OSPF, BGP, static) with PE — no VPN complexity on CE. Customer manages CE; provider manages PE and P.

Q190 What is VRF in Junos?

Ans VRF (Virtual Routing and Forwarding) creates separate, independent routing tables within a single router — used for MPLS L3VPN and network segmentation. Each VRF: has its own routing table, interfaces, and routing protocols. Routes from one VRF do not leak into another unless explicitly configured. Create VRF: set routing-instances CUSTOMER-A instance-type vrf. Add interfaces: set routing-instances CUSTOMER-A interface ge-0/0/1.0. Configure route distinguisher: set routing-instances CUSTOMER-A route-distinguisher 65001:100. Import/export policies control VPN routes.

Q191 What is MPLS traffic engineering?

Ans Traffic Engineering (TE) uses RSVP to direct traffic along specific paths through the network — not necessarily the shortest IGP path. Why? The shortest path may be congested while alternative paths have spare capacity. TE can: spread load across multiple paths, guarantee bandwidth for critical traffic (video, VoIP), provide fast reroute (protect against failures). Junos TE uses CSPF (Constrained Shortest Path First) to calculate paths considering bandwidth availability.

Q192 What is CSPF (Constrained Shortest Path First)?

Ans CSPF is the path calculation algorithm for MPLS traffic engineering. It runs on the head-end router and finds the shortest path that meets constraints: available bandwidth, link colour/affinity, explicit include/exclude paths. Uses the TE topology database (flooded by OSPF-TE or ISIS-TE with bandwidth info on each link). CSPF: finds path → RSVP sets up LSP along that path → bandwidth is reserved on each link. Without CSPF, RSVP would need explicit hop-by-hop path definition.

Q193 What is fast reroute (FRR) in MPLS?

Ans FRR provides sub-50 millisecond traffic restoration when an MPLS link or node fails. Pre-computed backup LSPs (bypass tunnels) are ready before failure. When failure detected: traffic immediately switched to backup – much faster than IGP convergence (which takes seconds). Types: Link protection (backup if link fails), Node protection (backup if entire router fails). Configure: set protocols mpls label-switched-path MY-LSP fast-reroute. Verify: show mpls label-switched-path detail (see protection status).

Q194 What is the difference between LDP and RSVP for MPLS?

Ans LDP: simple, automatic, follows IGP shortest path, no bandwidth reservation, no explicit path control, quick to deploy, scales easily. Best for: basic L2/L3 VPN services where exact path does not matter. RSVP: complex, requires explicit configuration or CSPF path calculation, enables bandwidth reservation, explicit path control, traffic engineering. Best for: guaranteed-bandwidth services, traffic engineering, large SP networks needing load control. Most providers use RSVP for core LSPs and LDP for L3VPN.

Q195 What is 6PE (IPv6 Provider Edge) in MPLS?

Ans 6PE allows IPv6 traffic to be carried over an IPv4 MPLS backbone – no need to upgrade the entire backbone to IPv6. How: PE routers are dual-stack (IPv4+IPv6). They use MP-BGP to exchange IPv6 routes with 'IPv6-specific' label encoding. IPv6 customer traffic gets labelled at ingress PE and forwarded over IPv4 MPLS backbone. Egress PE removes labels and routes IPv6 to customer. This is a cost-effective IPv6 migration strategy for service providers with large IPv4 MPLS networks.

Q196 What is MPLS Penultimate Hop Popping (PHP)?

Ans PHP removes the top MPLS label ONE hop before the egress router — the penultimate (second-to-last) router pops the label and forwards as IP (or with remaining VPN label). Why? The egress router does not need to do two lookups (label table then IP table) — it just does IP routing. Reduces processing on egress PE. PHP is enabled by default in Junos via the 'implicit null' label (label 3). PHP is transparent to customer traffic — purely internal MPLS optimisation.

Q197 What is RSVP bandwidth reservation in MPLS?

Ans RSVP LSPs can reserve bandwidth on each link along their path. When an RSVP LSP with 100 Mbps reservation is set up, the network deducts 100 Mbps from the available bandwidth on each traversed link. New LSPs requesting bandwidth are only routed via links with sufficient remaining capacity. This prevents over-subscription and ensures guaranteed performance for each LSP. View reservations: `show rsvp session extensive` (see bandwidth, path, status). `show rsvp neighbor` (see adjacent RSVP routers).

Q198 What is BFD (Bidirectional Forwarding Detection) in Junos?

Ans BFD detects link failures extremely quickly — in milliseconds — much faster than routing protocol hello timers. It sends lightweight probes between neighbours. Configure: `set protocols ospf area 0 interface ge-0/0/0.0 bfd-liveness-detection minimum-interval 300 multiplier 3`. This means: probe every 300ms, declare down after 3 missed probes (900ms total). Used with: OSPF, BGP, static routes, MPLS LSP monitoring. BFD is essential in networks requiring sub-second failover.

Q199 What is MPLS OAM (Operations, Administration, and Maintenance)?

Ans MPLS OAM tools verify and troubleshoot LSP connectivity. Junos tools: `ping mpls ldp fec 192.168.1.0/24` (ping along LDP LSP to verify connectivity). `ping mpls rsvp-te lsp-ping MY-LSP` (ping along RSVP LSP). `traceroute mpls ldp fec 192.168.1.0/24` (trace label path hop-by-hop, see labels swapped at each hop). These are MPLS-specific equivalents of IP ping/traceroute — essential for verifying LSP paths are working correctly end-to-end.

Q200 What is inter-AS MPLS VPN?

Ans Inter-AS VPN extends MPLS L3VPN connectivity between different Service Provider networks (different Autonomous Systems). Why? Large enterprises may need connectivity between sites served by different providers. Options: Option A (VRF-to-VRF at border): simplest, scalable but high state. Option B (ASBR-to-ASBR with labelled BGP): middle ground. Option C (multi-hop MP-eBGP for VPN routes): most scalable. In Junos: configured using AS override policies, multi-hop BGP, and route reflectors. Complex but necessary for multi-provider VPN deployments.

Q201 What is Juniper's PTX series in MPLS context?

Ans Juniper PTX series (P-series — Packet Transport Router) are high-capacity core routing platforms designed specifically for service provider backbone networks. Features: ultra-high port density (100GbE/400GbE), extremely high throughput (petabit-scale), optimised for IP/MPLS forwarding. PTX routers typically sit as P (Provider) routers in the MPLS core — pure label switching, no customer VRF tables. Junos runs on PTX — same CLI commands, just on much larger hardware.

Q202 What is the Junos MPLS label-switched-path configuration?

Ans Configure an RSVP LSP: set protocols mpls label-switched-path TO-SITE-B to 10.0.0.2 (destination router ID). Optional bandwidth: set protocols mpls label-switched-path TO-SITE-B bandwidth 50m. Optional explicit path: set protocols mpls path EXPLICIT-PATH hop 10.1.1.1 strict hop 10.2.2.1 strict. Apply path: set protocols mpls label-switched-path TO-SITE-B primary EXPLICIT-PATH. Enable RSVP on interfaces: set protocols rsvp interface ge-0/0/0.0. Verify: show mpls label-switched-path detail.

Q203 What is 'show mpls label-switched-path' in Junos?

Ans show mpls label-switched-path displays all configured MPLS LSPs and their status: LSP name, destination, state (Up/Dn), bandwidth allocated, path in use (primary/secondary), number of bytes/packets forwarded. show mpls label-switched-path detail — additional info: explicit path, RSVP session, labels used, retry timer if down. Useful for: verifying LSP is up, checking bandwidth reservation, confirming correct path is being used.

Q204 What is MPLS class of service?

Ans MPLS supports QoS through two mechanisms: EXP (Experimental) bits in the MPLS label (3 bits = 8 values, now called Traffic Class bits). PE router marks EXP bits based on IP DSCP. P routers use EXP bits for queuing decisions. MPLS-aware DiffServ: routers maintain separate queues per EXP value. This ensures: voice traffic (EXP 5) gets strict priority queue, video (EXP 4) gets guaranteed bandwidth, best-effort (EXP 0) gets remaining capacity – even in MPLS core where IP headers are hidden inside labels.

Q205 What is VPLS (Virtual Private LAN Service)?

Ans VPLS creates a distributed virtual Ethernet switch across an MPLS network. Multiple customer sites connect as if they share the same Ethernet LAN – same broadcast domain, same subnet. How: PE routers learn customer MAC addresses and build flooding tables. Traffic between sites is encapsulated with MPLS labels. VPLS use cases: connecting data centres at Layer 2, providing Ethernet leased line services, multi-site cloud connectivity. Configure: set protocols vpls MY-VPLS site CE1 interface ge-0/0/1.0.

Q206 What is EVPN (Ethernet VPN) in Junos?

Ans EVPN is the modern replacement for traditional VPLS – uses MP-BGP to distribute MAC/IP information between PE routers. Advantages over VPLS: better control plane (BGP instead of flooding), support for multihoming (connect CE to multiple PEs for redundancy), integrated L2 and L3 forwarding, better loop prevention, seamless MAC mobility (VM migration). EVPN is the standard for modern data centre and service provider networks. Junos QFX and MX support EVPN-VXLAN for data centre fabric.

Q207 What is VXLAN in Junos context?

Ans VXLAN (Virtual Extensible LAN) tunnels Layer 2 frames over a Layer 3 IP network – extending VLANs across routed boundaries. Like MPLS but uses UDP encapsulation (port 4789) – works over standard IP networks without MPLS. VXLAN ID (VNI) is 24-bit = 16 million segments (vs 4094 for 802.1Q). Used in: data centre fabrics (Juniper QFX with EVPN-VXLAN), cloud multi-tenant environments, extending VLANs between data centres. Junos supports VXLAN on QFX and MX series.

Q208 What is RPF (Reverse Path Forwarding) in Junos?

Ans RPF is a loop prevention mechanism for multicast routing. When a router receives a multicast packet, it checks: does the source IP address match a route in the unicast routing table that points back to the interface where the multicast packet arrived? If yes: accept (packet arrived from the correct direction). If no: discard (packet is arriving from wrong direction — loop or spoofed). RPF ensures multicast traffic flows in the correct direction. Also used for unicast RPF (uRPF) to prevent IP spoofing.

Q209 What is RSVP graceful restart in Junos?

Ans RSVP graceful restart allows RSVP LSPs to survive a Routing Engine restart (on dual-RE platforms) or RSVP process restart. During restart: RSVP neighbours are told to 'hold' LSP state. LSPs remain active in PFE (traffic continues forwarding). After restart: RE reconciles RSVP state with neighbours and resumes normal operation. No traffic interruption for existing LSPs. Configure: `set protocols rsvp graceful-restart`. Essential for high-availability service provider networks where LSP disruption is not acceptable.

Q210 What is MPLS ping and traceroute in Junos?

Ans MPLS OAM ping/traceroute verify LSP connectivity. Commands: `ping mpls ldp fec 192.168.1.1/32 count 5` (ping LDP LSP for this FEC). `ping mpls rsvp-te lsp-ping MY-LSP` (ping RSVP LSP). `traceroute mpls ldp fec 192.168.1.1/32` (trace label path, see each hop's IP and labels). Output shows: each hop, label values, MPLS TTL. Very useful for: verifying LSP is forwarding correctly, identifying exactly which hop has a problem, seeing label values at each point.

SECTION 08

High Availability & Network Management

Q211 –
Q240

VRRP · BFD · Graceful Restart · NSR · SNMP · NTP · Syslog · Automation

Q211 What is VRRP in Junos?

Ans VRRP (Virtual Router Redundancy Protocol) provides gateway redundancy – if the primary router fails, a backup takes over automatically. Configure: set interfaces ge-0/0/0 unit 0 family inet address 192.168.1.2/24 vrrp-group 1 virtual-address 192.168.1.1 priority 200 (this router wants to be master – highest priority wins). Backup router: same config but lower priority (default 100). Clients use 192.168.1.1 as gateway – they never know which physical router handles traffic. Verify: show vrrp detail.

TIP VRRP = *client never loses internet even if primary gateway router fails!*

Q212 How does VRRP master election work in Junos?

Ans VRRP master election: highest priority wins (set under vrrp-group). Priority range: 1-254 (default 100). If priorities are equal: highest IP address wins master role. The current IP address owner (router owning the virtual IP as a real interface IP) gets priority 255 – always wins. Preempt: if preempt is enabled (default in Junos), a router with higher priority takes master role when it recovers. Disable preempt: set interfaces... vrrp-group 1 no-preempt (current master stays master even if better router recovers).

Q213 What is BFD (Bidirectional Forwarding Detection) in detail?

Ans BFD is a fast detection protocol for link/node failures – operates in microseconds to milliseconds, much faster than routing protocol timers. How it works: BFD sessions between neighbours exchange periodic control packets (default 1 second). If packets stop (based on multiplier x interval): declare link down, immediately inform routing protocols. Configure with OSPF: set protocols ospf area 0 interface ge-0/0/0.0 bfd-liveness-detection minimum-interval 100 multiplier 5 (detect failure in 500ms). With BGP: set protocols bgp group IBGP bfd-liveness-detection minimum-interval 300.

Q214 What is GRES (Graceful Routing Engine Switchover) in Junos?

Ans GRES enables switchover from a failed primary Routing Engine to a backup RE on dual-RE platforms with zero traffic disruption. The backup RE stays in sync with the primary. When primary fails: backup becomes primary instantly. PFE continues forwarding. Protocols do not restart (requires NSR/GR to be configured too). GRES itself handles RE-level switchover. Combine with NSR for protocol continuity. Configure: `set chassis redundancy graceful-switchover`. Verify: `show chassis routing-engine` (see master/backup status).

Q215 What is NSR (Non-Stop Routing) in Junos?

Ans NSR maintains routing protocol sessions (BGP, OSPF, ISIS) during RE failover or routing daemon restart. The backup RE keeps live copies of all protocol states — neighbour tables, session state. When primary RE fails: backup instantly takes over with all protocol state intact. BGP neighbours see no session reset. OSPF neighbours see no adjacency break. Traffic forwarding continues uninterrupted. Configure: `set routing-options nonstop-routing`. Requires GRES to be enabled. Most important for BGP — session resets cause route withdrawal/re-advertisement which can disrupt traffic.

Q216 What is the difference between GRES, NSR, and Graceful Restart?

Ans GRES (Graceful Routing Engine Switchover): handles the RE hardware failover — zero traffic interruption during RE switchover on dual-RE platforms. NSR (Non-Stop Routing): maintains protocol session state during RE failover — neighbours do not see any protocol reset. Graceful Restart (GR): helper mode — neighbouring routers hold routes and do not withdraw them while a router restarts its routing daemon. GR works even without dual-RE hardware. Best practice: enable all three for maximum availability on carrier-grade platforms.

Q217 What is SNMP in Junos and how to configure it?

Ans SNMP (Simple Network Management Protocol) allows NMS (Network Management Station) to monitor Junos devices. Configure: `set snmp community PUBLIC authorization read-only`. `set snmp community SECRET authorization read-write`. `set snmp contact 'Akhilesh Chaudhari akhilesh@company.com'`. `set snmp location 'Data Center Floor 2'`. `set snmp trap-group MY-TRAPS targets 192.168.1.200` (NMS server IP). Enable traps for events: `set snmp trap-group MY-TRAPS categories link` (notify on interface up/down). SNMPv3 for security: `set snmp v3 usm local-engine user admin authentication-sha...`

Q218 What is NTP in Junos and how to configure it?

Ans NTP synchronises the device clock with a time server. Accurate time is critical for: log correlation (troubleshooting with logs from multiple devices), security certificates, Kerberos authentication. Configure NTP server: set system ntp server 216.239.35.0 prefer (Google NTP). set system ntp server 216.239.35.4. Manually set time zone: set system time-zone Asia/Kolkata. Verify: show ntp status (stratum, reference server, offset). show system uptime (current date/time). Juniper devices are NTP-server capable: set system ntp broadcast.

Q219 What is Syslog in Junos?

Ans Syslog sends log messages to a central server for storage and analysis. Configure: set system syslog host 192.168.1.100 any any (send all messages to syslog server). set system syslog host 192.168.1.100 port 514 (default UDP port). set system syslog file messages any notice (save notice-level and above to local file 'messages'). set system syslog file security-log authorization info (log authentication events). View local logs: show log messages. Real-time: monitor start messages. Junos syslog is very configurable — filter by facility and severity.

Q220 What is NETCONF in Junos automation?

Ans NETCONF (Network Configuration Protocol, RFC 6241) provides a programmatic interface to configure and monitor Junos devices using XML over SSH. Enables automation tools (Ansible, Python) to: retrieve configuration in structured XML, make configuration changes programmatically, subscribe to events. Enable: set system services netconf ssh. Test: ssh user@device -p 830 netconf (NETCONF uses port 830). Junos's NETCONF implementation is one of the best in the industry — very feature complete and reliable.

Q221 What is the Junos PyEZ library for automation?

Ans PyEZ is Juniper's Python library making Junos automation simple. Import: from jnpr.junos import Device. Connect: dev = Device(host='router', user='admin', password='pass'); dev.open(). Get facts: print(dev.facts['hostname']). Get config: cfg = dev.rpc.get_config(). Push config: from jnpr.junos.utils.config import Config; cu = Config(dev); cu.load('set system hostname TEST', format='set'); cu.commit(). Get operational data: result = dev.rpc.get_ospf_neighbor_information(). Much easier than raw NETCONF XML.

Q222 What is Junos commit scripting?

Ans Commit scripts run automatically every time 'commit' is executed – they can validate, transform, or add to the configuration. Uses: enforce configuration standards (if policy X exists, automatically add matching Y), prevent misconfigurations (refuse commit if required settings are missing), auto-generate configuration from templates. Scripts are written in SLAX or Python. Apply: set system scripts commit file MY-SCRIPT.slax. Very powerful for enforcing network-wide configuration standards automatically.

Q223 What is Junos op scripting?

Ans Op (operational) scripts run operational tasks and can be triggered from the Junos CLI. Create a custom Junos command: write a script that runs multiple show commands, formats output, and presents a summary. Example: type 'op network-health' and it automatically checks all interfaces, routing, and BGP status. Scripts in Python, SLAX, or shell. Install: copy to /var/db/scripts/op/. Enable: set system scripts op file MY-SCRIPT.py. Run: op MY-SCRIPT [arguments]. Great for standardising operational procedures.

Q224 What is Junos event scripting?

Ans Event scripts automatically run when specific system events occur – trigger-based automation. Example: when an interface goes down □ automatically: log a detailed message, execute 'show interfaces' to capture state, email the network team, and try a recovery command. Create event policy: set event-options policy IF-DOWN events SNMP_TRAP_LINK_DOWN then execute-commands commands 'show interfaces'. Apply script: set event-options policy IF-DOWN then event-script MY-SCRIPT.py. Enables self-healing network automation.

Q225 What is Ansible with Junos?

Ans Ansible automates Junos device configuration from a central control machine. Juniper provides official Ansible modules (juniper.device collection). Playbook example: use juniper.device.config module to push configurations to multiple devices simultaneously. Advantages: configure 100 devices in parallel, consistent configuration, version-controlled playbooks (Git), idempotent (safe to run multiple times), no software needed on Junos devices (uses NETCONF). Install: pip install ansible juniper.device. Practice in lab!

Q226 What is the Junos RESTCONF API?

Ans RESTCONF is an HTTP/HTTPS-based protocol for managing Junos using REST API calls — simpler to use than NETCONF (just HTTP requests). Enable: `set system services rest http (or https)`. Access device info: `GET https://device/rpc/get-interface-information`. Retrieve config: `GET https://device/restconf/data/configuration`. Modify config: `PATCH/PUT` requests with JSON or XML body. Authentication: basic auth or token-based. Great for: integration with IT ticketing systems, custom monitoring tools, web-based network automation.

Q227 What is YANG in Junos context?

Ans YANG (Yet Another Next Generation) is a data modelling language that defines the structure of Junos configuration and operational data — used with NETCONF and RESTCONF. Junos YANG models define: what configuration options exist and their types, what operational data (show commands) is available, relationships between configuration elements. Tools use YANG models to automatically know what data a device supports. Juniper publishes Junos YANG models on GitHub — use them with automation tools for accurate configuration generation.

Q228 What is Juniper's Contrail Networking?

Ans Contrail (now Juniper Apstra and Contrail Networking) is Juniper's SDN (Software Defined Networking) and cloud networking solution. Uses OpenContrail (open source) as base. Provides: automated network provisioning, intent-based networking (define what you want, Contrail figures out how), MPLS over VXLAN for virtual networks, micro-segmentation for cloud security. Used with OpenStack, Kubernetes, VMware. Juniper also offers Apstra: Intent-Based Networking platform for data centre automation.

Q229 What is Juniper Mist AI?

Ans Juniper Mist is an AI-driven cloud networking platform primarily for wireless networks (but expanding to wired and SD-WAN). Features: AI-powered wireless assurance (predicts and resolves Wi-Fi issues before users complain), virtual network assistant (Marvis AI bot — ask natural language questions about network health), cloud-managed APs, indoor location services, automated troubleshooting. Juniper acquired Mist Systems in 2019. Represents Juniper's AI-first networking strategy.

Q230 What is ZTP (Zero Touch Provisioning) in Junos?

Ans ZTP allows new Juniper devices to automatically configure themselves on first boot – no manual CLI needed. Process: device boots with factory default config, contacts DHCP server (gets IP and ZTP server info via DHCP options 66/43), downloads configuration or bootstrap script from ZTP server (HTTP), applies configuration automatically, device becomes operational. Benefits: deploy hundreds of devices without touching CLI. Tools: Juniper Automated WAN Solution, open-source ZTP servers. Very valuable for large-scale deployments.

Q231 What is Junos telemetry streaming?

Ans Junos supports streaming telemetry – devices continuously push performance data to collectors in real time (rather than polling via SNMP every 5 minutes). Protocol: gRPC with GNMI. Data: interface counters, CPU/memory, routing changes, BGP updates – at high frequency (every second). Benefits: detect brief traffic spikes that SNMP polling misses, real-time dashboards (Grafana), faster incident detection. Enable: set services analytics streaming-server MY-COLLECTOR. Subscribe to sensor: set services analytics export-profile ...

Q232 What is Junos chassis management?

Ans Chassis management provides hardware health monitoring and control. Commands: show chassis hardware (all installed components – FPCs, PICs, power supplies, fans). show chassis alarms (active hardware alarms). show chassis fpc (FPC status and temperature). show chassis environment (temperature, voltage, fan speed). request chassis fpc offline slot 3 (gracefully offline a line card). show chassis routing-engine (RE status, CPU, memory on dual-RE). These commands are essential for proactive hardware health monitoring.

Q233 What is Junos ISSU (In-Service Software Upgrade)?

Ans ISSU (In-Service Software Upgrade) upgrades Junos OS on high-end platforms (MX, PTX with dual RE) without any traffic disruption – zero downtime upgrades. Process: upgrade backup RE first, switchover traffic to backup RE, upgrade former primary RE, switchover back. Requires: dual REs, NSR+GRES enabled, ISSU-capable hardware. Not all Junos upgrades support ISSU – check release notes. Command: request system software in-service-upgrade [package] [options]. Verify: show system issu state.

Q234 What is RMON (Remote Monitoring) in Junos?

Ans RMON (Remote Monitoring, RFC 2819) extends SNMP with more advanced network monitoring capabilities. Junos supports RMON for: statistics (continuous interface traffic stats), history (periodic samples over time), alarms (threshold-based alerts), events (actions triggered by alarms). Configure alarm: `set snmp rmon alarm 1 variable ifInOctets.1 interval 60 rising-threshold 1000000 falling-threshold 500000`. RMON is less commonly used now (replaced by streaming telemetry) but still present in some enterprise environments.

Q235 What is the Junos system log hierarchy?

Ans Junos syslog has three destinations: Console (real-time on physical console – very verbose, often disabled). Terminal (specific user terminals). Files (stored locally – `/var/log/`). Configuration: Severity levels: emergency, alert, critical, error, warning, notice, info, debug (most verbose). Facility categories: any, authorization, daemon, ftp, interactive-commands, kernel, user, etc. Best practice: send severity notice and above to both local file and remote syslog server. Keep security-related logs (authorization) separate.

Q236 What is the Junos process hierarchy?

Ans Junos runs multiple processes (daemons) independently: `rpd` (Routing Protocol Daemon) – all routing protocols. `mgd` (Management Daemon) – CLI, NETCONF, J-Web. `chassisd` – hardware/chassis management. `dcd` (Device Configuration Daemon) – interface management. `mib2d` – SNMP. `sampled` – traffic sampling (sFlow). If one process crashes: Junos automatically restarts it without affecting others – microkernel architecture. View: `show system processes` (see all running processes and CPU usage).

Q237 What is Junos flow monitoring (sFlow/J-Flow)?

Ans Flow monitoring captures samples of network traffic for analysis. J-Flow (Cisco NetFlow equivalent): samples 1 in N packets, exports flow records (5-tuple: src/dst IP, ports, protocol) to a collector. Use for: bandwidth analysis, top talkers, security (detect DDoS, port scans). Enable: `set forwarding-options sampling input rate 1000 (sample 1 in 1000 packets). set forwarding-options sampling output cflowd 192.168.1.100 (export to collector)`. sFlow: alternative sampling protocol, Junos supports both.

Q238 What is the Junos core dump and how do you handle it?

Ans A core dump is a snapshot of a process's memory when it crashes — used for debugging. In Junos, if a daemon crashes (rpd, mgd, etc.), it creates a core dump file in `/var/tmp/`. Junos automatically restarts the crashed process. View core dumps: `show system core-dumps`. The device usually keeps running, but log files show 'process rpd has exited' type messages. Send core dumps to Juniper TAC for analysis when opening a support case. Clean up old cores: request system storage cleanup.

Q239 What is SNMP v3 configuration in Junos?

Ans SNMPv3 provides authentication and encryption — much more secure than v1/v2c community strings. Configure: `set snmp v3 usm local-engine user akhilesh authentication-sha authentication-password MyAuthPass privacy-aes128 privacy-password MyPrivPass`. Create access group: `set snmp v3 vacm security-to-group security-model usm security-name akhilesh group ADMIN-GROUP`. Set group access: `set snmp v3 vacm access group ADMIN-GROUP read-view all`. Verify: `show snmp v3` (see configured users and access).

Q240 What is the Junos help system?

Ans Junos has extensive built-in help: `?` — shows valid options at any point in a command. `help topic [keyword]` — explains a concept. `help reference [command]` — detailed command reference. `help apropos [keyword]` — finds commands related to a topic. `help syslog OSPF_NBR_STATE_CHNG` — explains what a syslog message means. Example: type 'set interfaces ?' and you see all valid interface configuration options. Very useful when you forget exact syntax — the CLI guides you without needing external documentation.

SECTION 09

Troubleshooting & Diagnostics

Q241 –
Q270*Methodology · Ping · Traceroute · Debug · Logging · Common Issues · Lab Tips***Q241** What is the Junos troubleshooting methodology?

Ans Systematic Junos troubleshooting: 1) Define the problem precisely (what is not working? since when?). 2) Check recent changes (show system commit – what changed and when?). 3) Start at Layer 1 (physical) and work up. 4) Use appropriate show commands at each layer. 5) Make ONE change at a time. 6) Verify after each change. 7) Document what you found and fixed. 8) Rollback if change makes things worse. The Junos commit model makes step 8 trivial – 'rollback 1; commit' instantly undoes last change.

TIP *Junos rollback makes troubleshooting safer – you can always undo quickly!*

Q242 How do you troubleshoot an interface that is down?

Ans Troubleshooting 'interface down': show interfaces ge-0/0/0 terse (confirm it shows 'down'). Check if 'admin disabled': show interfaces ge-0/0/0 – look for 'Administratively down'. Fix: delete interfaces ge-0/0/0 disable. Physical check: check cable (try replacing), check SFP (try another port or swap SFP), verify far end is active. Check errors: show interfaces ge-0/0/0 detail (CRC errors = bad cable). Check duplex/speed mismatch. If connected to a switch – check switch port status.

Q243 How do you troubleshoot OSPF neighbours not forming?

Ans OSPF neighbour troubleshooting: show ospf neighbor (see current state). Stuck in Init: far-end receives hellos but this end does not – check IP connectivity (ping neighbour IP). Stuck in 2-Way or ExStart: MTU mismatch – check both sides 'show interfaces' for MTU (must match). No neighbour at all: is OSPF enabled on correct interface? (show ospf interface). Area mismatch? (show ospf interface detail – check area). Authentication mismatch? (check auth key). check firewall filter not blocking OSPF protocol 89.

Q244 How do you troubleshoot BGP not establishing?

Ans BGP troubleshooting – show bgp summary: Active state = cannot connect to peer (TCP connection failing). Idle state = waiting to connect (admin or error). OpenSent/OpenConfirm = negotiating (authentication or capability mismatch). Established = working. Debug: show log messages | match BGP (check error messages). Verify: ping peer IP (basic connectivity). Check: TCP port 179 not blocked by firewall (show security flow session destination-port 179). AS number correct (both ends configured correctly). MD5 authentication key match.

Q245 How do you troubleshoot routing issues in Junos?

Ans Routing troubleshooting: show route [destination] (is there a route?). show route forwarding-table destination [destination] (did it make it to FIB?). traceroute [destination] (where does it fail?). ping [next-hop] (is Layer 3 working to next router?). show arp (ARP resolving correctly?). If route missing from FIB but in RIB: check if better-preference route exists from another protocol, check if route was rejected by policy. If route missing from RIB: routing protocol not running, policy filtering it, neighbour not sending it.

Q246 What is the Junos 'monitor traffic' command?

Ans monitor traffic interface ge-0/0/0 – captures live traffic on the interface and displays packet headers (like tcpdump). Shows: source/destination IP, protocol, TCP flags, ICMP type. Options: monitor traffic interface ge-0/0/0 count 10 (capture 10 packets). monitor traffic interface ge-0/0/0 matching 'host 192.168.1.5' (filter by IP). monitor traffic interface ge-0/0/0 extensive (detailed packet info). Very powerful for: seeing exactly what packets are arriving/leaving, verifying NAT is working, diagnosing application connectivity.

Q247 What is the Junos 'traceroute monitor' command?

Ans traceroute monitor [destination] continuously runs traceroute and updates the display – like the Linux 'mtr' command. Shows per-hop: IP address, packet loss percentage, average/min/max latency. Very useful for identifying where intermittent latency or packet loss is occurring in the path. Much better than running traceroute once – shows patterns over time. Press Ctrl+C to stop. Good for monitoring network path quality to ISP or cloud services during a suspected outage.

Q248 How do you use 'show route' to verify routing is correct?

Ans Verification workflow: show route [destination] – does a route exist? Check: route source (Direct, Static, OSPF, BGP), preference/metric values, next-hop IP and interface. show route [destination] detail – full attributes including AS-path, community, local-preference (BGP). show route forwarding-table [destination] – is it actually in FIB for forwarding? show route table [VRF] [destination] – check specific routing instance. If route exists but traffic fails – check ACLs/firewall policies. If route missing – debug the protocol.

Q249 What does 'request routing-engine login' do?

Ans On multi-RE platforms, 'request routing-engine login' lets you log into the backup RE from the primary RE console. Useful for: checking backup RE status, verifying GRES/NSR state on backup, troubleshooting RE failover issues. From backup RE prompt, run show commands to verify synchronisation. Type 'exit' to return to primary RE session. This is essential for maintaining dual-RE platforms – regularly verify backup RE is synchronised and healthy so failover works when needed.

Q250 What is the Junos 'test policy' command?

Ans test policy [policy-name] [route] evaluates a routing policy against a specific route prefix and shows what the outcome would be – without actually applying the policy. Very useful for: testing new policies before deploying, verifying existing policies work as expected, debugging why a route is being accepted or rejected. Output shows: which terms matched, what actions were taken, final result (accept/reject), any attribute modifications applied. Must be run from operational mode.

Q251 What is 'clear bgp neighbor' in Junos?

Ans clear bgp neighbor [IP] – resets the BGP session with a specific peer. Used when: BGP policy change needs to take effect (soft reset preferred), debugging connectivity, re-establishing stuck session. Options: clear bgp neighbor [IP] (hard reset – tears down and re-establishes TCP session and routing table exchange – may cause route flap). clear bgp neighbor [IP] soft (soft reset – re-runs import or export policy without tearing down session – preferred). Verify: show bgp summary (watch state go to Established).

Q252 What is 'show security flow session' for troubleshooting?

Ans show security flow session — shows all active firewall sessions on SRX. Each entry: source/destination IP, protocol, ports, security policy that permitted it, NAT translations. Filter: show security flow session source-prefix 192.168.1.5 (sessions from specific host). show security flow session destination-port 443 (only HTTPS sessions). If traffic should be flowing but is not: check if session is being created (is a policy permitting it?). No session = traffic being blocked. Verify which policy is matching: show security match-policies source-ip...

Q253 What are common Junos configuration mistakes?

Ans Frequent mistakes: Forgetting 'commit' (changes staged but not applied — looks broken but nothing actually changed). Wrong interface unit (configuring ge-0/0/0 but should be ge-0/0/0.0 in routing protocols). Family inet missing (interface has IP but 'family inet' not configured — no IPv4 processing). Zone assignment missing on SRX (interface not assigned to security zone — traffic silently dropped). Wrong wildcard mask in firewall filter. Not committing after rollback (rollback changes candidate config, still need commit).

Q254 How do you identify which policy is matching traffic on SRX?

Ans Use the security policy match command: show security match-policies from-zone trust to-zone untrust source-ip 192.168.1.5 destination-ip 8.8.8.8 destination-port 80 protocol tcp. This shows exactly which policy would match this traffic and what action would be taken. Essential for: verifying intended policy is actually matching, debugging why traffic is blocked or allowed unexpectedly, testing new policy logic before committing. Also: show security flow session — see policy name for active sessions.

Q255 What is 'show log' and how to use it effectively?

Ans show log messages — displays recent system log. Effective use: show log messages | match error — only errors. show log messages | match 'ge-0/0/0' — events for specific interface. show log messages | last 100 — last 100 lines. show log messages | count — count how many lines. show log messages | between [start] [end] (Junos supports time-based filtering). Real-time: monitor start messages — live stream. Stop: monitor stop messages. For SRX security events: show log security-log. For commit history with user: show system commit.

Q256 What is 'show class-of-service' for QoS troubleshooting?

Ans show class-of-service interface ge-0/0/0 — shows CoS configuration applied to interface: scheduler map, classifiers, rewrite rules. show class-of-service forwarding-class (configured forwarding classes — traffic categories). show interfaces ge-0/0/0 detail | match queue (per-queue counters — transmit, drop). For troubleshooting: if voice quality is poor — check voice queue has strict priority and not dropping packets (queue stats show drops = congestion). Adjust scheduler bandwidth allocations if dropping.

Q257 What is the 'no-more' pipe in Junos?

Ans show [command] | no-more — disables pagination and prints all output at once without waiting for you to press Space. Without no-more, large outputs pause at each page. Useful when: scripting or capturing output to a file (piping with no-more prevents interactive prompts), you want to scroll back through full output, copying output to notepad. Example: show bgp summary | no-more. Also useful in combination: show route | no-more | count (count total routes without paging).

Q258 What is 'show system processes extensive' in Junos?

Ans Displays all running processes with CPU and memory usage. Useful for: identifying high CPU (which process is consuming most CPU?), detecting crashed processes, monitoring daemon health. If rpd (routing daemon) using 100% CPU: OSPF/BGP instability. If mgd high CPU: many simultaneous CLI/API sessions. Process died: check log messages for 'exited' or 'core dumped'. Normal CPU: most daemons should be low (1-5%) in steady state. Spikes during commit or large routing table changes are normal.

Q259 How do you capture a complete diagnostic snapshot in Junos?

Ans request support information — collects comprehensive diagnostic snapshot: all show commands output, system logs, core dump list, hardware inventory. Saves to file: request support information | save /var/tmp/support.txt. Used when: opening a Juniper TAC case (attach this file). Gives TAC engineers complete picture without asking for individual outputs. Also: request system snapshot (copies OS + config to alternate media). File can be large (10-100 MB) — send via email or upload to TAC case. Very standard first step in any major issue.

Q260 What does 'show system storage' show in Junos?

Ans show system storage displays disk space usage for all Junos file system partitions: /var (logs, configs, software packages), /tmp, /boot, etc. Why it matters: if /var is full (100%), Junos cannot write logs, save configurations, or install software. Causes cryptic errors. Clean up: request system storage cleanup (removes old log files, rotated logs, unused software packages). Also manually remove large files: file delete /var/tmp/old-software.tgz. Best practice: monitor /var storage and keep below 80%.

Q261 What is traceroute with specific options in Junos?

Ans traceroute has useful options: traceroute 8.8.8.8 source 10.1.1.1 (use specific source IP — tests a specific interface or routing instance path). traceroute 8.8.8.8 routing-instance MGMT (test connectivity within a specific VRF). traceroute 8.8.8.8 ttl-exceeded (show TTL exceeded messages even if ICMP blocked). traceroute 8.8.8.8 no-resolve (skip DNS reverse lookup — faster output). traceroute 8.8.8.8 tos 40 (test with specific DSCP marking — verify QoS path).

Q262 What is the impact of 'commit full' in Junos?

Ans commit full — recompiles the entire configuration from scratch rather than applying only the changes (incremental commit). Use when: incremental commit is not working correctly, configuration database may be inconsistent, after certain types of complex changes. Slower than normal commit (may take 30-60 seconds on complex configs). Does not change what is applied — just how it is applied. Rarely needed but useful when you suspect commit database inconsistency. Always falls back to correct configuration.

Q263 How do you verify NAT translations in Junos SRX?

Ans show security nat source summary — overview of source NAT rules and hit counts. show security nat source rule all — all source NAT rules. show security nat source pool all — pool IP usage. show security flow session source-prefix 192.168.1.5 — active sessions with NAT translations visible. show security nat translation (older command) — all active NAT translations. If NAT not working: verify rule is being matched (check hit count), verify interface is correctly marked as 'inside' (from zone trust) and 'outside' (to zone untrust).

Q264 What is RMON alarm troubleshooting in Junos?

Ans If SNMP alerts not firing: show snmp rmon alarm (check threshold values and current variable value). show snmp rmon event (verify event action configured). show snmp statistics (verify SNMP is active). Check SNMP community string is correct. Verify NMS has correct community string. Test SNMP connectivity: from NMS, run 'snmpget -v2c -c PUBLIC device-IP sysDescr.0' – should return device description. Check firewall filter on lo0 not blocking UDP 161. show log messages | match SNMP (any SNMP errors in log).

Q265 What is the process for safe configuration changes in production?

Ans Safe change process in Junos: 1) save /var/tmp/pre-change-config.txt (backup). 2) show | compare (review all changes before commit). 3) commit check (validate syntax). 4) commit confirmed 10 (commit with 10-minute auto-rollback). 5) Test connectivity – verify change worked. 6) If OK: run 'commit' to confirm (cancel the auto-rollback timer). 7) If NOT OK: do nothing – auto-rollback triggers and restores original config in 10 minutes. 8) Document the change. This process makes production changes significantly safer.

Q266 What is 'show chassis alarms'?

Ans show chassis alarms – displays hardware alarms: Yellow alarm (minor issue – warning), Red alarm (major problem – needs immediate attention). Common alarms: FPC temperature too high, fan failure, power supply failure, SFP receive power too low, PIC offline. The physical device's alarm LEDs match these: yellow LED = yellow alarm, red LED = red alarm. Check chassis alarms daily as part of routine operations – catch hardware problems early before they cause outages.

Q267 How do you troubleshoot VLAN connectivity on EX switches?

Ans VLAN troubleshooting on EX: show vlans (does VLAN exist?). show interfaces ge-0/0/x (is port in correct VLAN?). show ethernet-switching interface ge-0/0/x (see VLAN assignment and learning state). show ethernet-switching table (is MAC address in table and on which port?). show interfaces irb.10 terse (is IRB gateway up?). ping 192.168.10.1 from client (can client reach gateway?). Trunk between switches: show interfaces [trunk-port] detail (see trunk allowed VLANs). Most VLAN issues: wrong VLAN assignment, trunk missing VLAN, IRB interface down.

Q268 What does show security policies hit-count show?

Ans show security policies hit-count – shows how many sessions each security policy has matched. Policies with zero hits may be: unnecessary (never triggered – safe to remove), misconfigured (traffic not matching as expected – policy below another policy catching it first), not yet triggered (new policy waiting for first match). Policies with very high hit counts are your busiest rules. This is very useful for: auditing security policies, cleaning up unused rules, verifying new policies are working after deployment.

Q269 What is 'monitor interface' for real-time stats?

Ans monitor interface [interface-name] – displays real-time interface statistics updating every second. Shows: input/output bytes per second, packets per second, errors. Press 'q' to quit. monitor interfaces (plural, no name) – all interfaces simultaneously in a table format – great for quick overall network health view. Useful for: seeing traffic spikes as they happen, verifying traffic is flowing through specific interface, monitoring bandwidth utilisation in real time during a maintenance window.

Q270 What are the most common Junos troubleshooting commands to memorise?

Ans Top 15 must-know Junos troubleshooting commands: show interfaces terse (all interface status). show route [prefix] (routing table lookup). show bgp summary (BGP peer status). show ospf neighbor (OSPF adjacency). show log messages | last 50 (recent system events). show security flow session (SRX active connections). show arp no-resolve (ARP table). show ethernet-switching table (MAC table on EX). show chassis alarms (hardware alarms). show system commit (recent changes). show vlan (VLAN config). ping/traceroute (connectivity test). show route forwarding-table (FIB). show system storage (disk space). show version (device info).

TIP

These 15 commands solve 80% of Junos network issues – memorise them all!

SECTION 10

JNCIA Career & Exam Preparation

Q271 –
Q300*Exam Tips · Study Strategy · Career Paths · JNCIS · Lab Practice · Industry***Q271** What is the JNCIA-Junos exam (JN0-105)?

Ans JNCIA-Junos (Juniper Networks Certified Internet Associate — Junos) is Juniper's entry-level certification for all networking tracks. Exam code: JN0-105. Format: 65 multiple-choice questions, 90 minutes. No prerequisites. Covers: Junos OS fundamentals, CLI, interfaces, routing basics, switching basics, network management. Passing score: approximately 70% (not officially published). Available at Pearson VUE testing centres globally. Valid for 3 years. Foundation for all Juniper specialist certifications.

TIP *JNCIA-Junos is the gateway to all Juniper career tracks — start here!*

Q272 What topics are covered in JNCIA-Junos (JN0-105)?

Ans Exam objectives: Junos OS Fundamentals (user interfaces, software architecture, CLI basics), Junos Configuration Basics (hierarchy, set/delete/commit, rollback, load/save), Interface Configuration (physical, logical, IPv4, IPv6, loopback), Routing Fundamentals (RIB/FIB, static routes, OSPF basics, BGP basics, routing policy), Switching (EX series, VLANs, STP/RSTP, IRB), and Network Management (SNMP, NTP, syslog, basic automation). All topics with equal importance.

Q273 What is the best way to study for JNCIA-Junos?

Ans Proven study approach: Official Juniper fast-track course (Junos Platform course — free on Juniper Learning Portal). Day1: Junos OS Essentials (free official Juniper eLearning). Practice in Juniper vLabs (free virtual Junos lab environments from Juniper). Use the free Junos vSRX or vMX on your laptop (VirtualBox/VMware) for hands-on. Read Juniper documentation. Take practice exams. Focus: CLI commands, commit model, rollback, routing policy structure. Connect Junos concepts to your existing IT knowledge.

TIP *Juniper's free learning portal (learningportal.juniper.net) is the best resource!*

Q274 What are the Juniper free learning resources?

Ans Free Juniper resources: Juniper Learning Portal (learningportal.juniper.net): free eLearning courses including Day1: Junos OS Essentials. Juniper Open Learning: free certification prep paths. Juniper vLabs: free temporary lab environments with real Junos devices (request a lab session). Juniper TechLibrary (documentation): complete Junos command reference. Juniper YouTube channel: video tutorials. Reddit [r/juniper](https://www.reddit.com/r/juniper): community Q&A.; NSM (Network Security Manager) training videos. All free – no excuses not to practice hands-on!

Q275 How is JNCIA-Junos different from Cisco CCNA?

Ans JNCIA-Junos focuses on Junos OS specifics: the commit model, rollback, configuration hierarchy, Junos CLI differences. Less emphasis on physical topologies and more on Junos architecture. CCNA covers a broader range of topics including WAN, automation, security at more depth. JNCIA is generally considered slightly narrower but deeper on Junos-specific concepts. Having CCNA first gives you routing/switching knowledge that transfers well to JNCIA – you will learn Junos syntax much faster. Many engineers have both.

Q276 What career opportunities does JNCIA-Junos open?

Ans JNCIA-Junos relevant roles: Junior Network Engineer (ISP, telco), Network Operations Centre (NOC) Engineer, Service Provider Network Technician, Managed Services Network Engineer. Salary range India: ₹4-8 LPA with JNCIA. ISPs and telecoms (Jio, Airtel, BSNL infrastructure teams), Juniper partner companies, large enterprises with Juniper networks. Many ISP networks in India and globally run on Juniper MX routers – JNCIA demonstrates you can work on them.

Q277 What is JNCIS-ENT and how does it build on JNCIA?

Ans JNCIS-ENT (Juniper Networks Certified Internet Specialist – Enterprise Routing and Switching) is the next level after JNCIA. Exam: JN0-351 (Enterprise Routing and Switching). Topics: Advanced OSPF (multi-area, redistribution, stub areas), Advanced BGP (attributes, policies, route reflection), IS-IS, advanced switching (RSTP, MSTP, LAG), L2 services, CoS, high availability (VRRP, BFD, GRES, NSR), and basic automation. Requires deeper understanding and more CLI practice than JNCIA. Same JNCIA-Junos exam is prerequisite for most tracks.

Q278 What are the different JNCIA tracks?

Ans JNCIA tracks (all start with JNCIA level): JNCIA-Junos (JN0-105): general Junos foundation – required first. JNCIA-SEC (JN0-230): security specialisation – SRX, VPN, IPS. JNCIA-DevOps (JN0-221): network automation – Ansible, Python, NETCONF. JNCIA-DC (JN0-211): data centre – QFX, EVPN, VXLAN. JNCIA-Cloud (JN0-211): cloud networking. JNCIA-Junos is foundation; other JNCIAs are specialisations. Most people start with JNCIA-Junos then specialise based on career direction.

Q279 How do you set up a Junos home lab for JNCIA practice?

Ans Free lab options: Juniper vSRX (free trial on AWS Marketplace – runs real Junos). Juniper vMX (trial version available). GNS3 or EVE-NG: run Junos VMs (need Junos qcow2 image – download from Juniper support portal with free account). Juniper vLabs: Juniper provides free temporary lab access for certification candidates. Minimum lab: 2 vSRX/vMX instances connected via virtual networks – enough to practice routing, BGP, OSPF, policies. PuTTY for SSH access. Wireshark for packet analysis.

Q280 What Junos commands should you know for the JNCIA exam?

Ans Critical CLI commands for JNCIA: show version, show interfaces terse, show route, show ospf neighbor, show bgp summary, show log messages, show system commit, show vlan (EX), show spanning-tree bridge/interface, show security policies (SRX), show security flow session (SRX), show arp, show ethernet-switching table (EX), rollback X, commit check, commit, commit confirmed, show | compare, request system reboot, show chassis hardware, ping, traceroute. Practise each of these until comfortable.

Q281 What are the common JNCIA exam question types?

Ans Typical question formats: Identify CLI command output (shown output from show command – what does it mean?). Select the correct configuration (which set command correctly achieves a goal?). Troubleshooting scenario (interface is down – which command would help diagnose?). Architecture understanding (what does the PFE do? Which plane handles routing protocols?). Sequence questions (what is the correct order of steps for an operation?). Best practice (which configuration is recommended?). No simulation/lab questions in JNCIA – all multiple choice.

Q282 What should you memorise for the JNCIA-Junos exam?

Ans Key facts to memorise: Interface naming (ge=1G, xe=10G, et=100G, lo0=loopback, fxp0=management). CLI modes (> operational, # configuration, edit/top/up navigation). Commit model (candidate $\square$ commit $\square$ active). Rollback stores last 50 configs. Route preference values (Direct=0, Static=5, OSPF=10, BGP=170 – different from Cisco!). vlan.dat is not a concept in Junos (VLAN config is in main config file). RSTP default on EX switches. IRB = SVI equivalent. family inet = IPv4 configuration.

Q283 What are the Juniper certification exam rules?

Ans Exam rules: Available at Pearson VUE test centres. Online proctoring option (verify current availability). Valid ID required (government issued). No scratch paper allowed (provided by testing centre). No notes, books, or phone access. English only (JNCIA-Junos). Exam is adaptive – questions may vary in difficulty. If you fail: retake policy requires 14 days wait before retaking. Register: pearsonvue.com/juniper. Cost approximately \$200 USD. Certification valid 3 years – recertify by passing higher-level exam or retaking same exam.

Q284 What is the Juniper certification recertification policy?

Ans Juniper certifications are valid for 3 years. To recertify: Pass the current version of the same exam (e.g. JN0-105 for JNCIA-Junos). Pass a higher-level exam in the same track (JNCIS-ENT recertifies JNCIA-Junos). Earn continuing education credits through approved Juniper training. Juniper sends reminder emails before expiration. Check status: Juniper's certification tracking portal. Important: if JNCIA expires, you lose foundation for JNCIS – keep it current. Many engineers renew by getting JNCIS.

Q285 What soft skills complement Junos technical knowledge?

Ans Technical skills + soft skills = successful network engineer: Documentation discipline (update network diagrams after every change), Structured troubleshooting (methodical approach, not random guessing), Communication (explain outages to management in non-technical terms), Collaboration (work with server/application/security teams), Change management (follow process – document, test, rollback plan), Continuous learning (networking evolves – stay current with Juniper releases, new features). These skills matter as much as CLI knowledge in a professional environment.

Q286 How does Junos knowledge compare in the job market to Cisco?

Ans Job market comparison: Cisco CCNA/CCNP are more widely required (Cisco has higher market share in enterprise). Juniper JNCIA/JNCIS is required/preferred for: ISP and telecom roles (Juniper dominates ISP market), financial sector (many banks use Juniper), data centre roles (Juniper QFX/MX widely deployed). Salary premium: Juniper-certified engineers often command higher salaries in SP sector — fewer certified engineers = higher demand. Best strategy: hold BOTH Cisco CCNA and Juniper JNCIA — broadest market coverage.

Q287 What is the value of Juniper JNCIA in India's IT market?

Ans In India, Juniper skills are particularly valued at: Telecom companies (Jio, Airtel, BSNL — large Juniper deployments), IT services companies with global clients needing Juniper support (TCS, Infosys, HCL), financial sector banks and stock exchanges (low-latency network requirements = Juniper MX/PTX), data centres (Juniper QFX), and MPLS backbone maintenance roles. With 4+ years experience and JNCIA, Senior Network Engineer roles at ISPs become accessible. Growing demand as 5G networks deploy Juniper equipment.

TIP

Your IT infrastructure experience at Chandan Steel is an excellent JNCIA foundation!

Q288 What is the difference between vMX and vSRX for lab practice?

Ans vMX: virtual version of MX series router. Best for practicing: routing protocols (OSPF, BGP, ISIS), MPLS, traffic engineering, routing policies, VRF. Looks and behaves like a real service provider router. vSRX: virtual version of SRX security gateway. Best for practicing: firewall zones, security policies, NAT, IPsec VPN, IPS. Both run full Junos OS — same CLI commands. For JNCIA-Junos exam: either works. For general routing knowledge: use vMX. For security concepts: use vSRX. Free trials available on cloud platforms (AWS, Azure, GCP).

Q289 What is Junos open community and support resources?

Ans Juniper community resources: Juniper Learning Portal (learningportal.juniper.net): free courses, certifications. Juniper Community Forum (community.juniper.net): Q&A, configuration examples, troubleshooting help. Juniper TechLibrary (tech.juniper.net): complete documentation, command reference, configuration guides. Reddit [r/juniper](https://www.reddit.com/r/juniper): active community. Twitter/LinkedIn: follow @JuniperNetworks for updates. GitHub: Juniper posts configuration examples and automation scripts. Juniper TAC (support.juniper.net): for support cases — requires valid support contract.

Q290 What are real-world scenarios where Junos knowledge is applied daily?

Ans Real-world Junos daily tasks: Adding a new office (configure WAN interface, BGP, static routes, commit). VLAN for new department (create VLAN, configure IRB, update security policies). BGP maintenance (soft reset BGP without dropping sessions, verify route changes). Firewall rule change (add security policy, test with show security match-policies, commit). Upgrade Junos OS (download, validate, install, reboot, verify). Troubleshoot outage (check show log, show route, show bgp summary, rollback if needed). All of these use the exact commands covered in JNCIA.

Q291 What is Juniper's Apstra platform?

Ans Juniper Apstra is an Intent-Based Networking platform for data centre automation. Engineers define WHAT they want (intents: 'I need a network connecting server rack A to storage cluster B') and Apstra determines HOW to configure it across Juniper (and other vendor) devices. Features: automated data centre fabric design, continuous compliance verification (ensures running config matches intent), automated troubleshooting, multi-vendor support (not just Juniper). Apstra is Juniper's answer to the complexity of large data centre network management.

Q292 What is Juniper's contribution to open networking?

Ans Juniper is active in open networking: OpenContrail (open-sourced Contrail SDN controller). Open19 (open server form factor standard). OpenROADM (open optical networking). Junos support for OpenConfig (vendor-neutral YANG models). YANG models published on GitHub (anyone can use them for automation). JNPRFly (open-source Ansible roles for Junos). Participation in IETF standards development (RSVP, MPLS, BGP standards). Juniper believes open networking reduces vendor lock-in and accelerates innovation – aligns with modern DevOps networking.

Q293 What is the Juniper Day One book series?

Ans Day One books are free, concise technical guides published by Juniper focusing on practical configuration examples. Topics: Day One: Configuring Junos Basics, Day One: Using OSPF on Junos, Day One: VPNs for Dummies, Day One: Data Centre Fundamentals. Download: pubs.juniper.net (free PDF). These books are excellent for exam preparation – they show real-world configurations with explanations. Written by Juniper engineers – authoritative and practical. Many JNCIA/JNCIS candidates use Day One books alongside official training materials.

Q294 What happens after JNCIA — what should I study next?

Ans After JNCIA-Junos, choose your specialisation: Enterprise Track: JNCIS-ENT (advanced OSPF, BGP, switching, HA) — best for enterprise network engineers. Service Provider Track: JNCIS-SP (advanced MPLS, BGP, L3VPN, L2VPN) — best for ISP/telecom careers. Security Track: JNCIA-SEC then JNCIS-SEC (SRX advanced, VPN, IPS) — best for security roles. DevOps Track: JNCIA-DevOps (automation, Python, Ansible) — best for modern network automation roles. Your IT background (networks, security, VPN) makes both Enterprise and Security tracks natural progressions.

Q295 What is Juniper's involvement in 5G networks?

Ans Juniper is a major 5G infrastructure provider: MX Series routers: 5G mobile packet core (5GC), user plane function (UPF). PTX Series: 5G transport network backbone. QFX Series: 5G data centre. Session Smart Router (SSR, acquired from 128 Technology): software-based routing for 5G edge and enterprise. Juniper 5G solutions enable: network slicing, ultra-low latency, massive device connectivity. With India's 5G rollout (Jio, Airtel using Juniper equipment), JNCIA/JNCIS engineers with 5G knowledge are increasingly valuable.

Q296 What is SR-MPLS (Segment Routing)?

Ans Segment Routing (SR) simplifies MPLS by encoding the complete path in the packet header — no per-flow state needed in intermediate routers (unlike RSVP which needs state at every hop). Each path segment is represented by a label. The first router (ingress) pushes a stack of labels encoding the entire path. Benefits: simpler operations (no RSVP), traffic engineering without per-flow state, scales better for large networks, easier automation. Junos supports SR-MPLS and SRv6 (Segment Routing over IPv6) — the future of SP networking.

Q297 What is SRv6 (Segment Routing over IPv6)?

Ans SRv6 implements Segment Routing using IPv6 addresses as segment identifiers instead of MPLS labels. No MPLS infrastructure needed — runs over standard IPv6. Each IPv6 address can represent a routing instruction. Benefits: simplified operations (no label management), native IPv6, easier to debug (trace with normal IPv6 tools), supports all MPLS services (L3VPN, L2VPN) but simpler. Junos MX and PTX support SRv6. The industry is moving toward SRv6 for new deployments — Cisco and Nokia also support it.

Q298 What is the future of Juniper in the networking industry?

Ans Juniper's strategic direction: AI-Driven Enterprise (Mist AI — self-driving networks for campus/branch/WAN). Cloud-ready networking (strong AWS/Azure/GCP integration). Open and Automated (NETCONF, YANG, Ansible, Python — excellent automation support). Security integration (network + security converging on SRX/Firefly). 5G infrastructure. Juniper's differentiation: best-in-class routing platform performance (MX series), AI networking (Mist acquisition), and strong service provider presence. Juniper skills will remain valuable as these directions grow.

Q299 How should you approach the JNCIA-Junos exam on exam day?

Ans Exam day strategy: Read every question completely before answering — Junos questions often have specific wording. Eliminate obviously wrong answers first. Pay attention to the platform (router vs switch vs SRX — different commands apply). Watch for 'MOST accurate', 'BEST option' — may be more than one correct but one is best. For CLI output questions: understand what each field means, not just the command. Manage time: 65 questions in 90 minutes = ~83 seconds per question. Mark unsure questions, answer all, review at end.

Q300 What final advice would you give to a JNCIA-Junos candidate?

Ans Professional advice from one network engineer to another: Understand the WHY behind Junos design decisions — the commit model, RE/PFE separation, hierarchy — they exist for excellent reasons. Hands-on practice in a lab (even virtual) is non-negotiable — reading alone is not enough. Connect JNCIA concepts to real-world problems you have already solved in your IT career. The commit model and rollback will feel natural once you have used them a few times. Start with JNCIA-Junos then add CCNA complement — the combination makes you extremely versatile. Network engineering is a career of continuous learning — JNCIA is a great starting point, not a finish line.

TIP

Akhilesh, your enterprise IT background (FortiGate, VPN, AD, network security) is a perfect foundation for JNCIA-Junos. You will find many familiar concepts with a Juniper flavour. Best of luck — you've got this!